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Antibody against claudin 18A2 and use thereof

Abstract

Provided are an anti-CLDN18.2 antibody or an antigen-binding fragment thereof, a derivative comprising said antibody or antigen-binding fragment thereof, a pharmaceutical composition, and related use of said antibody or antigen-binding fragment thereof for treating, diagnosing and detecting cancers.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS

(1) This application is a national stage application under 35 U.S.C. § 371 of International Application No. PCT/CN2020/090427, filed internationally on May 15, 2020, which claims priority to Chinese patent application No. 201910406762.4 filed May 16, 2019. The contents of the above patent applications are incorporated by reference herein in their entireties for all purposes.

SUBMISSION OF SEQUENCE LISTING ON ASCII TEXT FILE

(2) The content of the following submission on ASCII text file is incorporated herein by reference in its entirety: a computer readable form (CRF) of the Sequence Listing (file name: 166102000400SeqList.txt, date recorded: Nov. 15, 2021, size: 70,757 bytes).

TECHNICAL FIELD

(3) The present disclosure pertains to the field of immunology, and more particularly to antibodies

against Claudin 18.2 (CLDN18A2, CLDN18.2) or antigen-binding fragments thereof, derivatives comprising the antibodies or antigen-binding fragments thereof, pharmaceutical compositions, and related uses thereof in the treatment of cancer.

BACKGROUND

(4) Claudins are integral membrane proteins comprising a major structural proteins of tight junctions, such as the apical cell-cell adhesive junctions in polarized cell types seen in epithelial or endothelial cell layers. The tight junctions are composed of multi-strands reticular proteins that form a continuous seal around the cell, providing a physical barrier to solute and water transport in the paracellular space, but which is adjustable. The family of claudins contains at least 23 members in humans, ranging from 22 to 34 kDa in size. Although claudins are important for the function and stability of normal tissues, tumor cells often exhibit aberrant tight junction function. This may be associated with dysregulated expression and/or location of claudins due to dedifferentiation of tumor cells or the requirement for effective absorption of nutrients in tumor masses with abnormal angiogenesis by fast-growing cancer tissues (Morin, 2005, PMID: 16266975). Individual claudin family members may be up-regulated in certain cancer types, but down-regulated in other cancer types. Claudin 18 (CLDN18) is an integral membrane protein located in the tight junctions of epithelium and endothelium, with a molecular weight of about 27.9 KD. CLDN18 forms intercellular tight junctions with other tight junction proteins, regulating the permeability of tissue molecules and ions in the intercellular space, and maintaining the stability of the tissue environment. It is known that there are 2 subtypes of claudin 18, splice variant 1 (CLDN18A1, CLDN18.1): GenBank Accession Nos. NP_057453 and NM016369, and splice variant 2 (CLDN18A2, CLDN18.2): GenBank Accession Nos. NM_001002026 and NP_001002026. In normal cells, CLDN18A1 is selectively expressed in the epithelial cell of the lung, while CLDN18A2 is specifically expressed in normal gastric epithelial differentiated cells and not expressed in gastric epithelial stem cells with cell division activity. However, CLDN18A2 is overexpressed in tumor cells in many cancer types, such as high expression of CLDN18A2 found in 75% of gastric cancer patients, 50% of pancreatic cancer patients, and 30% of esophageal cancer patients, also in lung cancer and other cancer types. Therefore, finding antibodies that specifically bind to CLDN18A2 but not CLDN18A1 is of great significance for the treatment and detection of cancer. The existing CLDN18A2 antibody IMAB362 has entered the clinical research stage, clinical results showed that in gastric cancer patients with high expression of CLDN18.2 ($\geq 70\%$ of tumor cells with CLDN18.2 expression $\geq 2+$), compared with chemotherapy alone, the progression-free survival of chemotherapy+IMAB362 was extended from 6.1 months to 9.1 months, HR=0.46; the total survival time was extended from 9.3 months to 16.6 months, HR=0.44. In addition to antibody IMAB362, CAR-T prepared against CLDN18.2 targets has also entered clinical studies. However, these antibodies (or antigen-binding fragments in CAR-T) that have entered the clinical stage have less affinity to claudin 18.2. Therefore, there remains a need to continue to screen and prepare CLDN18.2 antibodies with higher affinity to produce greater potency at the same dosage.

SUMMARY OF THE INVENTION

(5) The present disclosure provides an anti-CLDN18.2 antibody or antigen-binding fragment thereof that specifically binds to CLDN18.2 and does not significantly bind to CLDN18.1.

(6) In some embodiments, the CLDN18.2 is a peptide having GenBank Accession No. NP_001002026 (mRNA:NM_001002026). The CLDN18.1 is a peptide having GenBank Accession No. NP_057453 (mRNA:NM_016369).

(7) In some embodiments, there is no significant binding between an antibody or antigen-binding fragment thereof of the present disclosure and CLDN18.1. In some examples, the antibody or antigen-binding fragment thereof binds to CLDN18.1 at a level of no more than 20% of that of binding to CLDN18.2. For example, the level of binding may be 20%, 19%, 18%, 17%, 16%, 15%, 14%, 13%, 12%, 11%, 10%, 9%, 8%, 7%, 6%, 5%, 4%, 3%, 2%, 1% or less than 1% of that of binding of the antibody or antigen-binding fragment thereof to CLDN18.2. In some embodiments,

the antibody or antigen-binding fragment thereof of the present disclosure binds to CLDN18.2 at a level 1-fold, 2-fold, 3-fold, 4-fold, 5-fold, 6-fold, 7-fold, 8-fold, 9-fold, 10-fold, or more than 10-fold greater than that of binding to CLDN18.1.

(8) The present disclosure provides an anti-CLDN18.2 antibody or antigen-binding fragment thereof, the antibody or antigen-binding fragment thereof is capable of specifically binding to CLDN18.2, comprising: a heavy chain variable region, the heavy chain variable region comprises three HCDRs selected from the group consisting of SEQ ID NOs: 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50, 51, 52, 53, 54, 55, 56, 57, 58, 59, 60, 61, 62, 63, 64, 65, 66, 67, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79, 80, 81, 82, 83, 84, 85; and/or, a light chain variable region, the light chain variable region comprises three LCDRs selected from the group consisting of SEQ ID NOs: 86, 87, 88, 89, 90, 91, 92, 93, 94, 95, 96, 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110, 111, 112, 113.

(9) In some embodiments, the present disclosure provides an anti-CLDN18.2 antibody or antigen-binding fragment thereof comprising a heavy chain variable region and/or a light chain variable region, the heavy chain variable region comprises HCDR1 selected from the group consisting of SEQ ID NOs: 37, 40, 43, 45, 49, 53, 56, 59, 62, 65, 68, 71, 74, 77, 80, 83; and HCDR2 selected from the group consisting of SEQ ID NOs: 38, 41, 46, 48, 50, 52, 54, 57, 60, 63, 66, 69, 72, 75, 78, 81, 84; and HCDR3 selected from the group consisting of SEQ ID NOs: 39, 42, 44, 47, 51, 55, 58, 61, 64, 67, 70, 73, 76, 79, 82, 85; and/or, the light chain variable region comprising LCDR1 selected from the group consisting of SEQ ID NOs: 86, 87, 88, 89, 90, 91, 92, 111, 112, 113; and LCDR2 selected from the group consisting of SEQ ID NOs: 93, 94, 95, 96; and LCDR3 selected from the group consisting of SEQ ID NOs: 97, 98, 99, 100, 101, 102, 103, 104, 105, 106, 107, 108, 109, 110.

(10) In a preferred embodiment, the present disclosure provides an anti-CLDN18.2 antibody or antigen-binding fragment thereof comprising a heavy chain variable region and/or a light chain variable region, the heavy chain variable region comprises HCDR1, HCDR2, and HCDR3 selected from the group consisting of: SEQ ID NO: 37, SEQ ID NO: 38 and SEQ ID NO: 39 or SEQ ID NO: 40, SEQ ID NO: 41 and SEQ ID NO: 42 of SEQ ID NO: 43, SEQ ID NO: 41 and SEQ ID NO: 44 or SEQ ID NO: 45, SEQ ID NO: 46 and SEQ ID NO: 47 or SEQ ID NO: 37, SEQ ID NO: 48 and SEQ ID NO: 39 or SEQ ID NO: 49, SEQ ID NO: 50 and SEQ ID NO: 51 or SEQ ID NO: 49, SEQ ID NO: 52 and SEQ ID NO: 51 or SEQ ID NO: 53, SEQ ID NO: 54 and SEQ ID NO: 55 or SEQ ID NO: 56, SEQ ID NO: 57 and SEQ ID NO: 58 or SEQ ID NO: 59, SEQ ID NO: 60 and SEQ ID NO: 61 or SEQ ID NO: 62, SEQ ID NO: 63 and SEQ ID NO: 64 of SEQ ID NO: 65, SEQ ID NO: 66 and SEQ ID NO: 67 or SEQ ID NO: 68, SEQ ID NO: 69 and SEQ ID NO: 70 or SEQ ID NO: 71, SEQ ID NO: 72 and SEQ ID NO: 73 or SEQ ID NO: 74, SEQ ID NO: 75 and SEQ ID NO: 76 or SEQ ID NO: 77, SEQ ID NO: 78 and SEQ ID NO: 79 or SEQ ID NO: 80, SEQ ID NO: 81 and SEQ ID NO: 82 of SEQ ID NO: 83, SEQ ID NO: 84 and SEQ ID NO: 85; and/or the light chain variable region comprises LCDR1, LCDR2, and LCDR3 selected from the group consisting of: SEQ ID NO: 86, SEQ ID NO: 93 and SEQ ID NO: 97, or SEQ ID NO: 87, SEQ ID NO: 94 and SEQ ID NO: 98; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 99; or SEQ ID NO: 87, SEQ ID NO: 95 and SEQ ID NO: 100; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 97; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 101; or SEQ ID NO: 89, SEQ ID NO: 93 and SEQ ID NO: 102; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 100; or SEQ ID NO: 90, SEQ ID NO: 93 and SEQ ID NO: 103; or SEQ ID NO: 91, SEQ ID NO: 96 and SEQ ID NO: 104; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 98; or SEQ ID NO: 92, SEQ ID NO: 93 and SEQ ID NO: 105; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 106; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 107; or SEQ ID NO: 87, SEQ ID NO: 93 and SEQ ID NO: 108; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 109; or SEQ ID NO: 88, SEQ ID NO: 93 and SEQ ID NO: 110; or SEQ ID NO: 111, SEQ ID NO: 93 and SEQ ID NO: 107; or SEQ ID NO: 112, SEQ ID NO: 93 and SEQ ID NO: 107; or SEQ ID NO: 113, SEQ ID NO: 93 and SEQ ID NO: 107;

or SEQ ID NO: 111, SEQ ID NO: 93 and SEQ ID NO: 110; or SEQ ID NO: 112, SEQ ID NO: 93 and SEQ ID NO: 110; or SEQ ID NO: 113, SEQ ID NO: 93 and SEQ ID NO: 110.

(11) According to one aspect of the present disclosure, the anti-CLDN18.2 antibody or antigen-binding fragment thereof comprises a heavy chain variable region and a light chain variable region, the variable region comprising 6 CDRs from any one of the following groups, the 6 CDRs of each group being arranged in order of HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, LCDR3: (1) SEQ ID NOs: 37, 38, 39, 86, 93, 97; (2) SEQ ID NOs: 40, 41, 42, 87, 94, 98; (3) SEQ ID NOs: 43, 41, 44, 88, 93, 99; (4) SEQ ID NOs: 45, 46, 47, 87, 95, 100; (5) SEQ ID NOs: 37, 48, 39, 88, 93, 97; (6) SEQ ID NOs: 49, 50, 51, 88, 93, 101; (7) SEQ ID NOs: 49, 52, 51, 89, 93, 102; (8) SEQ ID NOs: 53, 54, 55, 88, 93, 100; (9) SEQ ID NOs: 56, 57, 58, 90, 93, 103; (10) SEQ ID NOs: 59, 60, 61, 91, 96, 104; (11) SEQ ID NOs: 62, 63, 64, 88, 93, 98; (12) SEQ ID NOs: 65, 66, 67, 92, 93, 105; (13) SEQ ID NOs: 68, 69, 70, 88, 93, 106; (14) SEQ ID NOs: 71, 72, 73, 88, 93, 107; (15) SEQ ID NOs: 74, 75, 76, 88, 93, 106; (16) SEQ ID NOs: 77, 78, 79, 87, 93, 108; (17) SEQ ID NOs: 80, 81, 82, 88, 93, 109; (18) SEQ ID NOs: 83, 84, 85, 88, 93, 110; (19) SEQ ID NOs: 71, 72, 73, 111, 93, 107; (20) SEQ ID NOs: 71, 72, 73, 112, 93, 107; (21) SEQ ID NOs: 71, 72, 73, 113, 93, 107; (22) SEQ ID NOs: 83, 84, 85, 111, 93, 110; (23) SEQ ID NOs: 83, 84, 85, 112, 93, 110; (24) SEQ ID NOs: 83, 84, 85, 113, 93, 110.

(12) In some embodiments, the present disclosure provides an anti-CLDN18.2 antibody or antigen-binding fragment thereof comprising a heavy chain variable region and/or a light chain variable region, the heavy chain variable region has at least 80%, 85%, 90%, 95%, or 100% sequence identity to: SEQ ID NOs: 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35; and/or, the light chain variable region has at least 80%, 85%, 90%, 95%, or 100% sequence identity to: SEQ ID NOs: 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24, 26, 28, 30, 32, 34, 36, 114, 115, 116, 117, 118, 119.

(13) According to one aspect of the present disclosure, the anti-CLDN18.2 antibody or antigen-binding fragment thereof comprises at least 80% to 100% sequence identity to a heavy chain variable region and a light chain variable region of any one of the groups consisting of: (1) SEQ ID NOs: 1 and 2; (2) SEQ ID NOs: 3 and 4; (3) SEQ ID NOs: 5 and 6; (4) SEQ ID NOs: 7 and 8; (5) SEQ ID NOs: 9 and 10; (6) SEQ ID NOs: 11 and 12; (7) SEQ ID NOs: 13 and 14; (8) SEQ ID NOs: 15 and 16; (9) SEQ ID NOs: 17 and 18; (10) SEQ ID NOs: 19 and 20; (11) SEQ ID NOs: 21 and 22; (12) SEQ ID NOs: 23 and 24; (13) SEQ ID NOs: 25 and 26; (14) SEQ ID NOs: 27 and 28; (15) SEQ ID NOs: 29 and 30; (16) SEQ ID NOs: 31 and 32; (17) SEQ ID NOs: 33 and 34; (18) SEQ ID NOs: 35 and 36; (19) SEQ ID NOs: 27 and 114; (20) SEQ ID NOs: 27 and 115; (21) SEQ ID NOs: 27 and 116; (22) SEQ ID NOs: 35 and 117; (23) SEQ ID NOs: 35 and 118; (24) SEQ ID NOs: 35 and 119.

(14) In some preferred embodiments, the anti-CLDN18.2 antibody or antigen-binding fragment thereof described herein is a murine antibody, a chimeric antibody, or a humanized antibody.

(15) In some embodiments, the present disclosure provides an anti-CLDN18.2 antibody or antigen-binding fragment thereof comprising a heavy chain variable region and the light chain variable region, the heavy chain variable region has at least 80%, 85%, 90%, 95%, or 100% sequence identity to: SEQ ID NOs: 120, 122, 125, 128; and/or, the light chain variable region has at least 80%, 85%, 90%, 95%, or 100% sequence identity to: SEQ ID NOS: 121, 123, 124, 126, 127.

(16) According to one aspect of the present disclosure, the anti-CLDN18.2 antibody or antigen-binding fragment thereof comprises at least 80%, 85%, 90%, 95%, or 100% sequence identity to a heavy chain variable region and a light chain variable region of any one of the groups consisting of: (1) SEQ ID NOs: 120 and 121; (2) SEQ ID NOs: 120 and 123; (3) SEQ ID NOs: 120 and 124; (4) SEQ ID NOs: 122 and 121; (5) SEQ ID NOs: 125 and 126; (6) SEQ ID NOs: 125 and 127; (7) SEQ ID NOs: 128 and 126.

(17) In some preferred embodiments, the anti-CLDN18.2 antibody is a monoclonal antibody.

(18) In some preferred embodiments, the anti-CLDN18.2 antibody or antigen-binding fragment

thereof further comprises a heavy chain constant region and/or a light chain constant region, preferably the heavy chain constant region comprises an Fc or a variant Fc, and the Fc is derived from a mouse or a human.

(19) In some preferred embodiments, the anti-CLDN18.2 antibody is a full-length antibody.

(20) In some preferred embodiments, an anti-CLDN18.2 antibody or antigen-binding fragment thereof of the present disclosure is in the form of IgG1, IgG2, IgG3, or IgG4.

(21) In some preferred embodiments, the antigen-binding fragments of the present disclosure include Pab, Fv, scFv, F(ab').sub.2, linear antibodies, and single-domain antibodies.

(22) In some embodiments, the present disclosure provides a conjugate formed by coupling the afore-mentioned antibody or antigen-binding fragment thereof to a capture label or a detection label. Such detection labels include but are not limited to, radionuclides, luminescent substances (e.g., fluorescein), colored substances, or enzymes.

(23) In some embodiments, the present disclosure provides a bispecific or multispecific antibody, one antigen-binding domain of the bispecific or multispecific antibody comprises an anti-CLDN18.2 antibody or antigen-binding fragment thereof of the present disclosure.

(24) In some embodiments, the present disclosure provides an antibody-drug conjugate comprising an antibody or antigen-binding fragment thereof as previously described. The structure of such antibody-drug conjugates is well known in the art and is formed by the interconnection of antibody-linker-drug (toxin).

(25) In some embodiments, the present disclosure provides a chimeric antigen receptor in which an extracellular recognition unit comprises an antibody or antigen-binding fragment thereof as previously described.

(26) In some embodiments, the present disclosure provides a nucleic acid encoding any of the afore-mentioned antibodies or antigen-binding fragments thereof. According to another aspect of the present disclosure, there is provided a recombinant vector comprising the nucleic acid.

(27) In some embodiments, the present disclosure provides a host cell comprising the expression vector of the present disclosure or genome of which is integrated with the nucleic acid encoding the antibody or antigen-binding fragment thereof. In some preferred embodiments, the host cell may be a prokaryotic cell, such as *E. coli*; may also be eukaryotic cells such as yeast or mammalian cells such as CHO cells or HEK293 cells.

(28) In some embodiments, the present disclosure provides a method of preparing the antibody or antigen-binding fragment thereof, comprising: culturing the host cells of the present disclosure under suitable conditions and purifying the expression products from the cells.

(29) In some embodiments, the present disclosure provides the use of the antibody or antigen-binding fragment thereof for the preparation of a drug specifically targets CLDN18.2-expressing tumor cells, such as a monoclonal antibody drug, an antibody-drug conjugate, a bispecific antibody, or a multispecific antibody; or for the preparation of an immune cell modified by a chimeric antigen receptor; or for the preparation of a reagent for the diagnosis of a CLDN18.2-expressing tumor; in some embodiments, the CLDN18.2-expressing tumor comprises: gastric cancer, pancreatic cancer, esophageal cancer, lung cancer, ovarian cancer, colon cancer, liver cancer, head and neck cancer, and gallbladder cancer and metastases thereof, the gastric cancer metastasis such as Kuckenberg tumor.

(30) In some embodiments, the present disclosure provides a method of detecting CLDN18.2 expression in a sample, comprising: contacting the sample with the afore-mentioned anti-CLDN18.2 antibody or antigen-binding fragment thereof; detecting the formation of a complex of an anti-CLDN18.2 antibody or antigen-binding fragment thereof and CLDN18.2; optionally, the anti-CLDN18.2 antibody or antigen-binding fragment thereof is detectably labeled.

(31) In some embodiments, the present disclosure provides a pharmaceutical composition comprising an effective amount of an antibody or antigen-binding fragment thereof of the present disclosure, or comprising an effective amount of a nucleic acid encoding the antibody or antigen-

binding fragment thereof, or comprising an effective amount of a recombinant vector comprising a coding nucleic acid, or comprising an effective amount of a host cell comprising a coding nucleic acid, or comprising an effective amount of an antibody-drug conjugate of the present disclosure, or comprising an effective amount of a chimeric antigen receptor of the present disclosure, or comprising an effective amount of a bispecific or multispecific antibody of the present disclosure. In some embodiments, the pharmaceutical composition further comprises a pharmaceutically acceptable carrier.

(32) In some preferred embodiments, the pharmaceutical composition further comprises one or more additional therapeutic agents. Such additional therapeutic agents include cytotoxic agents, cytostatic agents, anti-angiogenic agents, anti-neoplastic agents, chemotherapeutic agents, radiotherapeutic agents, targeted anti-cancer agents, biological response modifiers, cancer vaccines, cytokines, hormones, anti-metastatic agents, and immunotherapeutic agents.

(33) In some embodiments, the present disclosure provides a drug box or a kit comprising a container and a pharmaceutical composition of the present disclosure in the container.

(34) In some embodiments, the present disclosure provides a method of inducing death in CLDN18.2-expressing cells, comprising contacting the cells with the pharmaceutical composition of the present disclosure. In some embodiments, the cells are contacted with the pharmaceutical composition in vitro. In some embodiments, the cells are contacted with the pharmaceutical composition in vivo. In some embodiments, the cell is a tumor cell.

(35) In some embodiments, the cell is a solid tumor cell. In some embodiments, the cell is selected from the group consisting of gastric cancer cells, esophageal cancer cells, intestinal cancer cells, pancreatic cancer cells, neuroblastoma cells, lung cancer cells, ovarian cancer cells, colon cancer cells, rectal cancer cells, liver cancer cells, head and neck cancer cells, chronic myelogenous leukemia cells, and gallbladder cancer cells.

(36) In some embodiments, the present disclosure provides a method of treating a disease associated with expression of CLDN18.2 in a subject, comprising administering to a subject in need thereof a pharmaceutical composition of the present disclosure. In some embodiments, the disease is a tumor. In some embodiments, the tumor is preferably gastric cancer, esophageal cancer, intestinal cancer, pancreatic cancer, neuroblastoma, lung cancer, ovarian cancer, colon cancer, rectal cancer, liver cancer, head and neck cancer, chronic myelogenous leukemia, or gallbladder cancer. In some embodiments, the method further comprises administering to the subject an additional therapeutic agent.

(37) The antibodies of the present disclosure may be administered in combination with another additional therapeutic agent, including, but not limited to, chemotherapeutic agents, cytotoxic agents, radiotherapeutic agents, cancer vaccines, anti-neoplastic agents, targeted anti-cancer agents, anti-angiogenic agents, biological response modifiers, cytokines, hormones, anti-metastatic agents, and immunotherapeutic agents.

(38) In some preferred embodiments, the chemotherapeutic agents that can be used in combination with an antibody or antigen-binding fragment thereof of the present disclosure include but are not limited to, mitotic inhibitors, including vincristine, vinblastine, vindesine, and navelbine; topoisomerase I inhibitors, such as camptothecin compounds, including irinotecan, topotecan and other compounds derived from camptothecin and analogs thereof; podophyllotoxin derivatives such as etoposide, teniposide, and midoxizox; alkylating agents such as cisplatin, carboplatin, cyclophosphamide, nitrogen mustard, trimethylenethiophosphoramide, carmustine, busulfan, chlorambucil, briquinolizine, uracil mustard, cloprofen, and dacarbazine; antimetabolites, including cytarabine, 5-fluorouracil, methotrexate, mercaptopurine, azathioprine, and procarbazine; antibiotics including, but not limited to, doxorubicin, bleomycin, dactinomycin, daunorubicin, mitomycin, sarcomycin C, actinomycin D, roxithromycin, adriamycin, rapamycin and derivatives thereof, and daunomycin; and other chemotherapeutic agents including, but not limited to, paclitaxel, docetaxel, dacarbazine, azacytidine, amsacon melphalan, ifosfamide, and mitoxantrone.

In some preferred embodiments, the additional therapeutic agent is selected from one or more of epirubicin, oxaliplatin, and 5-fluorouracil.

(39) In some embodiments, the targeted anticancer agents include but are not limited to, large molecule-targeted drugs, small molecule-targeted drugs, etc.

(40) In some preferred embodiments, the macromolecular targeting agents include but are not limited to, epidermal growth factor-targeted agents, including cetuximab, panitumumab, and nimotuzumab, etc.; HER-2 or HER-3 signaling pathway inhibitors, including trastuzumab, pertuzumab, T-DM1, etc.; anti-vascular endothelial growth factor drugs, including VEGF-TRAP, bevacizumab, ramucirumab, etc.; also, agents targeting other targets include but are not limited to, targets such as PI3K, PARP, PI3K α , PKB/AKT, and STAT3.

(41) In some embodiments, small molecule-targeted agents include but are not limited to, epidermal growth factor targeting agents, including erlotinib or gefitinib, etc.; HER-2 or HER-3 signaling pathway inhibitors, including lapatinib or afatinib, etc.; tyrosine kinase inhibitors including imatinib or sunitinib, etc.; anti-vascular endothelial growth factor drugs including sorafenib, regorafenib, pazopanib, recombinant human endostatin, apatinib, etc.; targeting c-Met/ROS1 drugs, including crizotinib, etc.; and, other targeting agents, including but not limited to vorinostat and marimastat, etc.; targeting mTOR drugs, including everolimus, etc.; and agents targeting other targets including but not limited to PI3K α , PKB/AKT, and STAT3.

(42) In some embodiments, the immunotherapeutic agents include but are not limited to, immunosuppressive agents and agonists, wherein the targets include PD-1/PD-L1, PD-L2, CTLA-4, LAG-3, IDO, TIM3, TIGIT, CD47, SIRP α , 4-1BB, CSF-1/CSF1R, GITR, OX40, CD40, CD27, CD28, B7H4, B7H3, TGF β , BTLA, VISTA, ICOS, CD39, CD73, A2AR, KIR, and NKG2A, etc.; and cell therapy associated with immunotherapy.

(43) In some embodiments, immune checkpoint inhibitors that target PD-1/PD-L1 include but are not limited to, macromolecular drugs such as, Pembrolizumab, Nivolumab, Atezolizumab, Avelumab, Sintilimab, Cemiplimab, and Durvalumab, etc.; and small molecule drugs.

(44) In some embodiments, immune checkpoint inhibitors that target CTLA-4 include but are not limited to, Ipilimumab, etc.; cytokines include but are not limited to, IL-10, IL-15, IL4, and IL13, etc.; inhibitors that target BRAF include but are not limited to, Binimetinib, etc.

(45) In some embodiments, the other therapeutic agent is selected from oncolytic viruses, such as parvovirus, adenovirus, herpes virus, poxvirus, poliovirus, reovirus, alphavirus, maraba virus, retrovirus, and coxsackie virus, etc.; alternatively, the other therapeutic agent is selected from cancer vaccines or protease inhibitors, such as bortezomib, etc.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) The drawings further illustrate the novel features disclosed in this specification. The features and advantages disclosed in this specification will be better understood with reference to the drawings, but it is to be understood that these drawings are merely illustrative of specific embodiments of the principles disclosed herein and are not intended to limit the scope of the appended claims.

(2) FIGS. 1-5 show the binding of supernatants from 18 hybridoma cell lines of the present disclosure to HEK293 cells stably transformed with hCLDN18.2 as measured by flow cytometry.

(3) FIGS. 6-11 show the binding of 18 chimeric antibodies of the present disclosure to HEK293 cells stably transfected expressing hCLDN18.2.

(4) FIGS. 12-17 show the binding of 18 chimeric antibodies of the present disclosure to gastric cancer tumor tissue-derived cells that naturally express hCLDN18.2.

(5) FIGS. 18-23 show ADCC results of 18 chimeric antibodies of the present disclosure against

CHO-K1 cells stably transfected expressing hCLDN18.2.

(6) FIGS. 24-29 show CDC results of 18 chimeric antibodies of the present disclosure against CHO-K1 cells stably transfected expressing hCLDN18.2.

(7) FIG. 30 shows the binding of the humanized antibody hu299B2-S32A of the present disclosure to HEK293 cells stably transfected expressing hCLDN18.2.

(8) FIG. 31 shows the binding of the humanized antibody hu253C4-N31Q of the present disclosure to HEK293 cells stably transfected expressing hCLDN18.2.

(9) FIG. 32 shows the binding of the humanized antibody hu299B2-S32A of the present disclosure to gastric cancer tumor tissue-derived cells naturally expressing hCLDN18.2.

(10) FIG. 33 shows the binding of the humanized antibody hu253C4-N31Q of the present disclosure to gastric cancer tumor tissue-derived cells naturally expressing hCLDN18.2.

(11) FIG. 34 shows ADCC results of humanized antibody hu299B2-S32A of the present disclosure against CHO-K1 cells stably transfected expressing hCLDN18.2.

(12) FIG. 35 shows ADCC results of humanized antibody hu253C4-N31Q of the present disclosure against CHO-K1 cells stably transfected expressing hCLDN18.2.

(13) FIG. 36 shows the CDC results of humanized antibody hu299B2-S32A of the present disclosure against CHO-K1 cells stably transfected expressing hCLDN18.2.

(14) FIG. 37 shows the CDC results of humanized antibody hu253C4-N31Q of the present disclosure against CHO-K1 cells stably transfected expressing hCLDN18.2.

DETAILED DESCRIPTION OF THE INVENTION

Terms

(15) All publications, patents, and patent applications mentioned in this specification are herein incorporated by reference to the same extent as if each publication, patent, or patent application was specifically and individually indicated to be incorporated by reference.

(16) Before the present disclosure is described in detail below, it is to be understood that the present disclosure is not limited to the particular methodology, protocols, and reagents described herein, as these may vary. It is also to be understood that the terms used herein are to describe particular embodiments only, and are not intended to limit the scope of the present disclosure. Unless otherwise specified, all technical and scientific terms used herein have the same meanings as those generally understood by a person skilled in the art to which the present disclosure belongs.

(17) Certain embodiments disclosed herein encompass numerical ranges, and certain aspects of the present disclosure may be described in terms of ranges. Unless otherwise indicated, it is to be understood that numerical ranges or descriptions of ranges are merely for brevity and convenience and should not be construed as strictly limiting the scope of the present disclosure. Accordingly, the description in a range format should be taken to specifically disclose all possible subranges and all possible specific numerical points within that range, as such subranges and numerical points are expressly written herein. For example, a description of a range from 1 to 6 should be considered to specifically disclose subranges from 1 to 3, 1 to 4, 1 to 5, 2 to 4, 2 to 6, 3 to 6, etc., as well as specific numerical points within these ranges, e.g., 1, 2, 3, 4, 5, 6. The above principles are equally applicable regardless of the width of the numerical values. Where a range description is employed, the range includes the endpoints of the range.

(18) The term “about” when referring to a measurable value such as an amount and temporal duration, refers to a change that includes $\pm 20\%$, or in some cases $\pm 10\%$, or in some cases $\pm 5\%$, or in some cases $\pm 1\%$, or in some cases $\pm 0.1\%$ of the specified value.

(19) Amino acid three-letter codes and one-letter codes as used herein are as described in J. Biol. Chem, 243, p3558(1968).

(20) As used herein, the terms “antibody of anti-Claudin18.2”, “anti-CLDN18.2 antibody”, or “antibody against CLDN18.2” refers to such an antibody that is capable of binding to the CLDN18.2 protein or fragment thereof with sufficient affinity such that the antibody can be used as a diagnostic and/or therapeutic agent that targets CLDN18.2. The human-derived CLDN18.2

protein is designated hCLDN18.2, thus, “antibody of anti-human Claudin18.2”, “anti-human Claudin18.2 antibody”, “antibody of anti-hCLDN18.2”, “anti-hCLDN18.2 antibody”, “antibody against hCLDN18.2” in particular to refers to such an antibody that is capable of binding to the human CLDN18.2 protein or fragment thereof with sufficient affinity such that the antibody can be used as a diagnostic and/or therapeutic agent that targets human CLDN18.2.

(21) The term “antibody”, as used herein, typically refers to a Y-type tetrameric protein comprising two heavy (H) polypeptide chains and two light (L) polypeptide chains held together by covalent disulfide bonds and non-covalent interactions. Native IgG antibodies have this structure. Each light chain consists of one variable domain (VL) and one constant domain (CL). Each heavy chain comprises one variable domain (VH) and a constant region.

(22) As known in the art, the heavy chain constant domains may be classified into α , δ , ϵ , γ and μ which define isotypes of an antibody as IgA, IgD, IgE, IgG, and IgM, respectively; and IgG and IgA may be further classified into different subclasses, wherein IgG may be subdivided into for example IgG1, IgG2, IgG3, and IgG4, and IgA may be subdivided into IgA1 and IgA2. The light chains of antibodies from any vertebrate species can be assigned to one of two distinct types, called κ and λ , based on the amino acid sequences of their constant domain. In IgG, IgA, and IgD, the constant region comprises three domains called CH1, CH2, CH3 (IgM and IgE have the fourth domain called CH4). In IgG, IgA, and IgD, the CH1 and CH2 domains are separated by a flexible hinge region, which is a proline-rich and cysteine-rich segment of variable length. Each type of antibody further comprises interchain and intrachain disulfide bonds formed by paired cysteine residues.

(23) The term “variable region” or “variable domain” shows a significant change in amino acid composition from one antibody to another and is primarily responsible for antigen recognition and binding. The variable region of each light/heavy chain pair forms an antibody binding site such that the intact IgG antibody has two binding sites (i.e., it is bivalent). Variable region of heavy chain (VH) and variable region of light chain (VL) each comprise three regions of extreme variability, which are termed hypervariable regions (HVR), or more generally, complementarity-determining regions (CDRs), VH and VL each have four framework regions (FRs), which are represented by FR1, FR2, FR3, and FR4, respectively. Thus, CDR and FR sequences typically occur in the following sequences of variable region of heavy chain (VH) (or variable region of light chain (VL)): FR1-HCDR1(LCDR1)-FR2-HCDR2(LCDR2)-FR3-HCDR3(LCDR3)-FR4.

(24) The term “Fc” is used herein to define the C-terminal region of an immunoglobulin heavy chain comprising at least a portion of a constant region. This term includes native sequence Fc regions and variant Fc regions. Unless otherwise indicated, the numbering of amino acid residues in the Fc region or constant region is according to the EU numbering system, which is also referred to as the EU index, as described in Kabat et al., Sequences of Proteins of Immunological Interest, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, MD, 1991.

(25) As used herein, the types of “antibodies” in a broad sense may include, for example, polyclonal antibodies, monoclonal antibodies, chimeric antibodies, humanized and primatized antibodies, CDR-grafted antibodies, human antibodies (including recombinantly produced human antibodies), recombinantly produced antibodies, intracellular antibodies, multispecific antibodies, bispecific antibodies, monovalent antibodies, multivalent antibodies, anti-idiotypic antibodies, synthetic antibodies (including muteins and variants thereof), etc.

(26) The terms “full-length antibody” and “intact antibody” are used interchangeably herein to refer to an antibody having a structure substantially similar to that of a native antibody structure or having an Fc region.

(27) The term “monoclonal antibody” (or “mAb”) refers to a substantially homogeneous antibody produced by a single cell clone that is directed against only a particular antigenic epitope. Monoclonal antibodies can be prepared using a variety of techniques known in the art, including hybridoma techniques, recombinant techniques, phage display techniques, transgenic animals,

synthetic techniques, or combinations thereof.

(28) The term “chimeric antibody” is a construct in which a portion of the heavy and/or light chain is identical or homologous to a corresponding sequence in an antibody from a particular species or belonging to a particular antibody class or subclass, and the remaining portion of the chain(s) is identical or homologous to a corresponding sequence in an antibody from another species or belonging to another antibody class or subclass, and corresponding sequences in fragments of such antibodies. In a narrow sense, a chimeric antibody comprises all or most of selected murine heavy and light chain variable regions operably linked to human light and heavy chain constant regions. The constant region sequences, or variants or derivatives thereof, may be operatively associated with the disclosed heavy and light chain variable regions using standard molecular biology techniques to provide full-length anti-CLDN18.2 antibodies that may be used themselves or may be incorporated into the present disclosure.

(29) The term “humanized antibody” is a hybrid immunoglobulin, immunoglobulin chain or fragment thereof that contains the smallest sequence derived from non-human immunoglobulin. In most cases, the humanized antibody is a human immunoglobulin (recipient antibody) in which residues from CDRs of the recipient are replaced by residues from CDRs of a non-human species (donor antibody) having the desired specificity, affinity, and properties, such as mice, rats, rabbits or primates. In some cases, the framework residues of a human immunoglobulin are replaced with corresponding non-human residues. In some cases, “back mutations” may be introduced into humanized antibodies in which residues in one or more FRs of the variable region of the recipient human antibody are replaced with corresponding residues from a non-human species donor antibody. Such back mutations may help maintain the proper three-dimensional configuration of one or more grafted CDRs, thus improving affinity and antibody stability. Antibodies from a variety of donor species including, but not limited to, mice, rats, rabbits, or non-human primates may be used. In addition, humanized antibodies may contain new residues not found in the recipient antibody or the donor antibody to further improve antibody performance.

(30) It is noted that the divisions of CDR and FR in the variable regions of the monoclonal antibodies of the present disclosure are determined according to the Kabat definition. However, other naming and numbering systems, such as Chothia, IMGT, or AHo, are also known to those skilled in the art. Thus, humanized antibodies comprising one or more CDRs derived from any nomenclature system, based on the monoclonal antibody sequences of the present disclosure, are expressly maintained within the scope of the present disclosure.

(31) The term “sequence identity” or “sequence similarity” or “sequence homology” refers to the percentage of amino acid residues in a candidate sequence that is identical to the same amino acid residues in a reference polypeptide sequence after the sequences are aligned (and gaps are introduced when necessary) to achieve the maximum percent sequence identity, and any conservative substitutions are not considered as part of the sequence identity.

(32) Sequence alignments can be performed using various approaches in the art to determine percent amino acid sequence identity, for example, using publicly available computer software such as BLAST, BLAST-2, ALIGN, or MEGALIGN (DNASTAR) software. Those skilled in the art can determine the appropriate parameters for the measurement alignment, including any algorithm required to achieve the maximum alignment over the full length of the sequence being compared.

(33) The term “antibody fragment” encompasses at least a portion of an intact antibody. As used herein, a “fragment” of an antibody molecule includes an “antigen-binding fragment” of an antibody, and the term “antigen-binding fragment” refers to a polypeptide fragment of an immunoglobulin or antibody that specifically binds to or reacts with a selected antigen or antigenic epitope thereof, or a fusion protein product further derived from the fragment, e.g., a single-chain antibody, an extracellular binding region in a chimeric antigen receptor, etc. Exemplary antibody fragments or antigen-binding fragments thereof include but are not limited to light chain variable fragments (VL), heavy chain variable fragments (VH), Fab fragments, F(ab').sub.2 fragments, Fd

fragments, Fv fragments, single domain antibodies, linear antibodies, single-chain antibodies (scFv), and bispecific antibodies or multispecific antibodies formed from antibody fragments, etc. (34) The term “Fab fragment” includes a variable region of each of the heavy and the light chain, and also includes a constant region of the light chain and a first constant region CH1 of the heavy chain, which is a monovalent antibody fragment. The term “F(ab’).sub.2 fragment” encompasses two Fab fragments as well as hinge regions, which is a bivalent antibody fragment.

(35) The term “Fd fragment” generally encompasses a heavy chain variable region and a constant region CH1; the term “Fv fragment” is the smallest antibody fragment having variable regions of heavy chain and light chain, but no constant region, and holding all antigen-binding sites.

(36) The term “scFv” refers to a fusion protein comprising at least one antibody fragment including the variable region of the light chain and at least one antibody fragment including the variable region of the heavy chain, wherein the variable regions of the light and heavy chain are connected (e.g., via a synthetic linker such as a short flexible polypeptide linker) and capable of being expressed as a single-chain polypeptide, and wherein the scFv retains the specificity of the intact antibody from which it is derived. Unless otherwise specified, an scFv may have the VL and VH variable regions described in any order (e.g., relative to the N-terminus and C-terminus of the polypeptide), and scFv may comprise a VL-linker-VH or may comprise a VH-linker-VL.

(37) The term “multispecific antibody” refers to a novel antibody construct binding to more than two different sites and/or targets, which is formed by functionally linking (e.g., chemical coupling, gene fusion, non-covalent binding, or other methods) the antibody to one or more other binding molecules. The more common multispecific antibody is the “bispecific antibody”, which specifically refers to an antibody construct with specificities for two different antigens. Typically, a bispecific or multispecific antibody comprises at least two antigen-binding domains.

(38) The term “antigen” refers to a substance recognized and specifically bound by an antibody or antibody-binding fragment, and broadly, an antigen can include any immunogenic fragment or determinant of a selected target, including a single epitope, a multi-epitope, a single domain, multiple domains, or an entire extracellular domain (ECD) or a protein. Peptides, proteins, glycoproteins, polysaccharides and lipids, portions thereof, and combinations thereof may constitute antigens. Non-limiting exemplary antigens include tumor antigens or pathogen antigens, etc. “Antigen” may also refer to a molecule that triggers an immune response. Any form of antigens or cells or preparations containing the antigens may be used to generate antibodies specific for an antigenic determinant. The antigen can be an isolated full-length protein, a cell surface protein (e.g., immunized with a cell expressing at least a portion of the antigen on its surface), or a soluble protein (e.g., immunized with only the ECD portion of the protein), or a protein construct (e.g., an Fc antigen). The antigen may be produced in genetically modified cells. Any of the foregoing antigens may be used alone or in combination with one or more immunogenicity-enhancing adjuvants known in the art. The DNA encoding the antigen may be genomic or non-genomic (e.g., cDNA) and may encode at least a portion of the ECD sufficient to trigger an immunogenic response. Any vector may be used to transform cells in which the antigen is expressed, including but not limited to adenoviral vectors, lentiviral vectors, plasmids, and non-viral vectors such as cationic lipids.

(39) The term “epitope” refers to a site on an antigen that specifically binds to an immunoglobulin or antibody. Epitopes may be formed from contiguous amino acids or noncontiguous amino acids juxtaposed by the tertiary folding of a protein. Epitopes formed from contiguous amino acids are typically retained upon exposure to denaturing solvents, while epitopes formed by tertiary folding are typically lost upon treatment with denaturing solvents. Epitopes typically comprise at least 3-15 amino acids in a unique spatial conformation. Methods for determining the epitope to which a given antibody binds are well known in the art, including immunoblotting and immunoprecipitation detection assays. Methods for determining the spatial conformation of an epitope include techniques in the art and described herein, such as X-ray crystallography, two-dimensional nuclear

magnetic resonance, etc.

(40) The terms “polypeptide”, “peptide”, and “protein” are used interchangeably herein to refer to polymers of amino acids of any length. The polymer may be linear, cyclic or branched, and may comprise modified amino acids, particularly conservatively modified amino acids, and it may be interrupted by non-amino acids. The term also includes modified amino acid polymers such as amino acid polymers that have been modified by sulfation, glycosylation, lipidation, acetylation, phosphorylation, iodination, methylation, oxidation, proteolytic processing, prenylation, racemization, selenoylation, transfer RNA (tRNA)-mediated amino addition such as arginylation, ubiquitination, or any other operation such as conjugation to a labeling component. As used herein, the term “amino acid” refers to natural and/or non-natural or synthetic amino acids, including glycine and D or L optical isomers, as well as amino acid analogs and peptidomimetics. A polypeptide or amino acid sequence “derived from” a given protein refers to the source of the polypeptide. The term also includes polypeptides expressed from the specified nucleic acid sequences.

(41) The term “amino acid modification” (or “modified amino acid”) includes amino acid substitutions, insertions, and/or deletions in a polypeptide sequence. As used herein, “amino acid substitution” or “substitution” refers to the replacement of an amino acid at a particular position in a parent polypeptide sequence with another amino acid. For example, substitution S32A means that serine at position 32 is replaced with alanine.

(42) Sequence identity or homology of a humanized antibody variable region to a human receptor variable region can be determined as discussed herein, and when measured in this way, the two will preferably share at least 60% or 65% sequence identity, more preferably at least 70%, 75%, 80%, 85% or 90% sequence identity, even more preferably at least 93%, 95%, 98% or 99% sequence identity. Preferably, residue positions that are not identical differ by conservative amino acid substitutions. A “conservative substitution” is an amino acid substitution in which one amino acid residue is replaced with another amino acid residue having a side chain (R group) with similar chemical properties (e.g., charge or hydrophobicity). In general, conservative amino acid substitutions do not substantially alter the functional properties of the protein. Families of amino acid residues having similar side chains have been defined in the art. These families include amino acids containing basic side chains (e.g., lysine, arginine, histidine), acidic side chains (e.g., aspartic acid, glutamic acid), uncharged polarside chains (e.g., glycine, asparagine, serine, threonine, tyrosine, cysteine, tryptophan), non-polar side chains (e.g., alanine, valine, leucine, isoleucine, proline, phenylalanine, methionine), B branched side chains (e.g., threonine, valine, isoleucine), and aromatic side chains (e.g., tyrosine, phenylalanine, tryptophan, histidine). Thus, one or more amino acid residues in the CDR regions or the framework regions of the antibodies of the present disclosure may be replaced with amino acid residues of other similar side chains. In the case where two or more amino acid sequences differ from one another by conservative substitutions, the percent sequence identity or degree of similarity may be adjusted upward to correct for the conservative nature of the substitution.

(43) In the production of monoclonal antibodies, various post-translational modifications (PTM) variants, such as glycosylation, oxidation, saccharification, deamidation, isomerization, and end-group cyclization, are easily produced by different physical and chemical factors. These PTMs may cause changes in the physical and chemical properties of the antibody, alter the interaction with the Fc receptor of the antibody, and affect the binding activity with the target antigen; the occurrence of some PTMs may even reduce antibody stability, cause immunogenicity, etc. (JARASCH et al., JOURNAL OF PHARMACEUTICAL SCIENCES, 2015). Negative effects can be eliminated by amino acid modifications, such as conservative substitutions, to the PTM site.

(44) Amino acid substitutions to antibody CDRs to modify PTM also expressly maintained within the scope of the present disclosure.

(45) The term “antibody-dependent cell-mediated cytotoxicity” (ADCC) refers to the binding of an

antibody to an epitope of a virus-infected cell or tumor cell, wherein Fc fragment binds to Fc receptors (FcRs) present on killer cells (NK cells, and macrophages, etc.) to mediate the killer cells to directly kill target cells.

(46) The term “complement-dependent cytotoxicity” (CDC) refers to the cytotoxic effect in the presence of complement, i.e. the lysis of target cells by membrane attack complex formed by activation of the classical complement pathway, which is initiated by complex formed by the binding of specific antibodies to corresponding membrane surface antigens.

(47) Antibodies of the present disclosure may also include substitutions or modifications of constant regions (e.g., Fc), including, but not limited to, amino acid residue substitutions, mutations, and/or modifications, which result in compounds having the following preferred characteristics, including, but not limited to: altered pharmacokinetics, increased serum half-life, increased binding affinity, decreased immunogenicity, increased yield, altered Pc ligand binding to Pc receptors (FcRs), increased or decreased ADCC or CDC, altered glycosylation and/or disulfide bonds, and modified binding specificity.

(48) The term “affinity” or “binding affinity” refers to the strength of the sum of all non-covalent interactions between a single binding site of a molecule (e.g., an antibody) and its binding partner (e.g., an antigen). The term “K_{sub}D” refers to the dissociation constant of a particular antibody-antigen interaction. Binding affinities can be determined using various techniques known in the art, such as surface plasmon resonance, bio-layer interferometry, dual-polarization interferometry, static light scattering, dynamic light scattering, isothermal titration calorimetry, ELISA, analytical ultracentrifugation, and flow cytometry, etc.

(49) The term “competitive binding” or “competitive antibody” generally refers to an antibody that binds to the same epitope as the antibody of the present disclosure, the binding of which results in the binding of the antibody of the present disclosure to the epitope to be inhibited or blocked, and the degree of competitive inhibition can be obtained in a competition assay.

(50) The term “pharmaceutical composition” refers to a formulation that is present in a form that allows the biological activity of the active ingredients contained therein to be effective, and which does not contain additional ingredients having unacceptable toxicity to the subject to which the formulation is administered.

(51) The term “pharmaceutically carrier” or “pharmaceutically acceptable carrier” refers to a diluent, adjuvant (e.g., Freund's adjuvant (complete and incomplete)), excipient, or vehicle with which a therapeutic agent is administered.

(52) The term “effective amount” refers to a dose of a pharmaceutical formulation of an antibody or fragment of the present disclosure that, when administered to a patient in single or multiple doses, produces the desired effect in the treated patient. An effective amount can be readily determined by the attending physician, as one skilled in the art, by considering the following factors: for example, the different of human species; body weight, age, and health; specific diseases involved; the severity of the disease; the response of an individual patient; the specific antibody administered; modes of administration; bioavailability characteristics of the administered formulation; a selected dosing regimen; and the use of any concomitant therapy.

(53) The terms “host cell”, “host cell line” and “host cell culture” are used interchangeably and refer to a cell into which an exogenous nucleic acid is introduced, including progeny of such a cell. Host cells include “transformants” and “transformed cells”, which include primarily transformed cells and progeny derived therefrom, regardless of the number of passages. The progeny may not be exactly the same as the parent cell in nucleic acid content but may contain mutations. Mutant progeny having the same function or biological activity as screened or selected in the initially transformed cell are included herein.

(54) As used herein, the term “transfection” refers to the introduction of an exogenous nucleic acid into a eukaryotic cell. Transfection can be accomplished by various means known in the art, including calcium phosphate-DNA co-precipitation, DEAE-dextran mediated transfection,

polybrene-mediated transfection, electroporation, microinjection, liposome fusion, lipid transfection, protoplast fusion, retroviral infection, and biolistics.

(55) The term “stable transfection” or “ST” refers to the introduction and integration of an exogenous nucleic acid, DNA, or RNA into the genome of a transfected cell. The term “stable transfectant” refers to a cell that stably integrates foreign DNA into genomic DNA.

(56) The terms “nucleic acid molecule encoding”, “coding DNA sequence” and “coding DNA” refer to the order of deoxyribonucleotides along a strand of deoxyribonucleic acid. The order of these deoxyribonucleotides determines the order of the amino acids along the polypeptide (protein) chain. Thus, the nucleic acid sequence encodes an amino acid sequence.

(57) Methods for producing and purifying antibodies and antigen-binding fragments are well known in the art and can be found, for example, in chapters 5-8 and 15 in *Using Antibodies: A Laboratory Manual*, Cold Spring Harbor Laboratory. The antibodies or antigen-binding fragments thereof of the present disclosure are genetically engineered to add one or more human FR regions to CDR regions of non-human origin. Human FR germline sequences can be obtained from the website imgt.cines.fr of ImMunoGeneTics (IMGT), or *J. Immunoglobulin*, (2001) ISBN: 012441351.

(58) The engineered antibodies or antigen-binding fragments thereof of the present disclosure can be prepared and purified by conventional methods. For example, cDNA sequences encoding heavy and light chains can be cloned and recombined into expression vectors. The recombinant immunoglobulin expression vector can stably transfect CHO cells. As a more recommended prior art, mammalian expression systems may result in glycosylation of antibodies, particularly at the highly conserved N-terminus of the Fc region. Stable clones are obtained by expressing antibodies that specifically bind to human antigens. Positive clones are enlarged cultured in a serum-free medium in a bioreactor to produce antibodies. The antibody-secreting medium may be purified and collected using conventional techniques. The antibody may be concentrated by filtration using conventional methods. Soluble mixtures and polymers may also be removed by conventional methods, such as molecular sieves, ion exchange, etc.

(59) As used herein, the term “individual” or “subject” refers to any animal, such as a mammal or a bagged animal. Individuals of the present disclosure include but are not limited to, humans, non-human primates (e.g., cynomolgus or rhesus monkeys or other types of macaque), mice, pigs, horses, donkeys, cattle, sheep, rats, and any kind of poultry.

(60) The term “antibody-drug conjugate” (ADC) refers to an antibody to which a therapeutically active substance or active drug ingredient (API) has been covalently coupled such that the therapeutically active substance or active drug ingredient (API) can be targeted to the binding target of the antibody to exhibit its pharmacological function.

(61) The therapeutically active substance or active pharmaceutical ingredient may be a cytotoxin capable of killing the ADC-targeted cells, preferably a malignant or cancerous cell. Covalent attachment of the therapeutically active substance, active pharmaceutical ingredient, or cytotoxin can be carried out in a non-site-specific manner using standard chemical linkers that couple the payload to lysine or cysteine residues, or preferably, conjugation is carried out in a site-specific manner that allows full control of the conjugation site and the drug to antibody ratio of produced ADC.

(62) The term “chimeric antigen receptor” (CAR) is an engineered receptor that transplants any specificity onto immune effector cells. Typically, these recipients are used to transplant the specificity of monoclonal antibodies onto T cells; the transfer of their coding sequences is facilitated by retroviral or lentiviral vectors or by transposons. CAR-engineered T cells (also abbreviated as CAR-T cells) are genetically engineered T cells harboring a chimeric receptor, the extracellular recognition unit of which comprises an antibody-derived recognition domain, and the intracellular region of which is derived from a lymphocyte stimulating moiety. The structure of the prototype CAR is modular and designed to accommodate various functional domains, thus

enabling the selection of specificity and control of T cell activation. The preferred antibody-derived recognition unit is a single-chain variable fragment (scFv) that combines the specificity and binding residues of the heavy and light chain variable regions of a monoclonal antibody. The most common lymphocyte activation moiety comprises a T-cell costimulatory (e.g., CD28) domain in tandem with a T-cell triggering (e.g., CD3C) moiety. By arming effector lymphocytes (e.g., T cells and natural killer cells) with such chimeric receptors, the engineered cell is redirected with a predefined specificity to any desired target antigen, in a non-HLA-restricted manner.

(63) CAR constructs are introduced ex vivo into T cells from peripheral lymphocytes of a given patient using retroviral or lentiviral vectors or transposons. Following infusion of the resulting CAR-engineered T cells back into the patient, they traffic, reach their target site, and upon interaction with their target cells or tissues, they undergo activation and perform their predefined effector function. Therapeutic targets for the CAR approach include cancer and HIV-infected cells, or autoimmune effector cells.

(64) As used herein, the term “tumor” refers to a disease characterized by pathological proliferation of cells or tissues, and subsequent migration or invasion of other tissues or organs. The growth of a tumor is usually uncontrolled and progressive and does not induce or inhibit normal cell proliferation. Tumors can affect various cells, tissues or organs, including, but not limited to, bladder, bone, brain, breast, cartilage, glial cells, esophagus, fallopian tube, gallbladder, heart, intestine, kidney, liver, lung, lymph nodes, nerve tissue, ovary, pancreas, prostate, skeletal muscle, skin, spinal cord, spleen, stomach, testis, thymus, thyroid, trachea, urethra, ureter, urethra, uterus, vaginal organ, or tissue or corresponding cell. Tumors include cancers, such as sarcomas, carcinomas, or plasmacytomas (malignant tumors of plasma cells). The tumor according to the present disclosure may include, but is not limited to, leukemia (e.g. acute leukemia, acute lymphocytic leukemia, acute myeloid leukemia, acute myeloid leukemia, acute promyelocytic leukemia, acute myelo-monocytic leukemia, acute monocytic leukemia, chronic leukemia, chronic myelogenous leukemia, chronic lymphocytic leukemia, polycythemia vera), lymphoma (Hodgkin's disease, non-Hodgkin's disease), primary macroglobulinemia, heavy chain disease, solid tumors such as sarcomas and cancers (e.g. fibrosarcoma, myxosarcoma, liposarcoma, chondrosarcoma, osteosarcoma, chordoma, endothelial sarcoma, lymphangiosarcoma, angiosarcoma, lymphangioendothelioma, mesothelioma, Ewing's tumor, leiomyosarcoma, rhabdomyosarcoma, colon cancer, pancreatic cancer, breast cancer, ovarian cancer, prostate cancer, squamous cell carcinoma, basal cell carcinoma, adenocarcinoma, sweat adenocarcinoma, sebaceous adenocarcinoma, papillary carcinoma, papillary adenocarcinoma, bronchial carcinoma, myeloid cancer, renal cell carcinoma, liver cancer, bile duct carcinoma, choriocarcinoma, seminoma, embryo cancer, nephroblastoma, cervical cancer, uterine cancer, testicular cancer, lung cancer, small cell lung cancer, bladder cancer, epithelial cancer, glioma, astrocytoma, medulloblastoma, craniopharyngioma, ependymoma, pinealoma, hemangioblastoma, acoustic neuromas, oligodendroglioma, schwannoma, meningioma, melanoma, neuroblastoma, retinoblastoma), esophageal cancer, gallbladder cancer, kidney cancer, multiple myeloma. Preferably, the “tumor” includes, but is not limited to: pancreatic cancer, liver cancer, lung cancer, gastric cancer, esophageal cancer, head and neck squamous cell carcinoma, prostate cancer, colon cancer, breast cancer, lymphoma, gallbladder cancer, renal cancer, leukemia, multiple myeloma, ovarian cancer, cervical cancer, and glioma.

(65) As used herein, the term “disease” or “condition” or “disorder” or the like refers to any alteration or disorder that impairs or interferes with the normal function of a cell, tissue, or organ. For example, the “disease” includes, but is not limited to: tumors, pathogen infections, autoimmune diseases, T-cell dysfunctions, or deficiencies in immune tolerance (e.g., transplant rejection).

(66) As used herein, the term “treatment” refers to clinical intervention in an attempt to alter a disease caused by an individual or treated cells, either prophylactically or clinically pathologically. Therapeutic effects include but are not limited to, prevention of the occurrence or recurrence of a

disease, alleviation of symptoms, reduction of any disease's direct or indirect pathological consequences, prevention of metastasis, slowing of the rate of disease progression, amelioration or remission of a condition, remission or amelioration of a prognosis, etc.

(67) The term “drug box” or “kit” includes an effective amount of one or more unit dosage forms of a pharmaceutical composition of the present disclosure. In some embodiments, the drug box may include a sterile container; such containers may be in the form of boxes, ampoules, bottles, vials, tubes, bags, blister packs, or other suitable containers known in the art. Such containers may be made of plastic, glass, laminated paper, metal foil, or other materials suitable for holding medicaments. In addition, the drug box also includes instructions for administering the pharmaceutical composition of the present disclosure to an individual. The instructions generally include methods of using the pharmaceutical compositions of the present disclosure to treat diseases.

EXAMPLE

(68) The present disclosure will be described in detail below in connection with specific examples. It should be understood that these examples are only used to describe the present disclosure and are not intended to limit the scope of the present disclosure. The experimental methods in the following examples which are not specified with specific conditions are generally carried out according to conventional conditions, *Molecular Cloning: A Laboratory Manual* (Third Edition) by J. Sambrook et al., Science Press, 2002, or according to the conditions recommended by the manufacturer.

Example 1 Animal Immunization

(69) To generate anti-CLDN18.2 antibodies, the assay procedure was performed using standard biological protocols.

(70) A total of 15 mice of different strains were immunized with CHO-K1 cells (CHO-K1/hCLDN18.2) stably transfected expressing hCLDN18.2 and a DNA vector encoding hCLDN18.2 as immunogens.

(71) In the later stage of immunization, blood was collected from the angular vein to obtain plasma samples, and the titer of immune serum was detected by ELISA and FACS to determine the immune response of the animals.

(72) After 4 times of immunization, 6 mice were selected for euthanasia to prepare hybridoma cells.

Example 2 Production of Hybridoma Cell for Anti-CLDN18.2 Monoclonal Antibodies

(73) To generate hybridoma cells for the anti-CLDN18.2 monoclonal antibody, 6 mice were euthanized with carbon dioxide, and feeder cells were separately harvested by syringe, and the feeder cell suspension was plated into a prepared 96-well plate. A certain number of myeloma cells and spleen cells were proportionally mixed for cell fusion. HAT medium (1 mL 100×HT supplement+1 mL aminopterin+10 mL FBS+88 mL DMEM) was added to the fused cells and mixed well to make cell suspension. The cell suspension was then poured into a culture dish and mixed well, and the cell suspension was plated into a 96-well feeder plate using a multichannel pipette. The fused feeder cell plates were placed in an incubator and incubated at a constant temperature of 37° C., 5.5% CO₂ for 7-10 days. Anti-CLDN18.2 positive clones were then screened by ELISA and FACS. The screened positive clones were subcloned by limiting dilution assay to obtain stable single hybridoma cells. Subcloned cell supernatants were screened by FACS using HEK293 cells stably transfected expressing hCLDN18.2 (HEK293-hCLDN18.2). As shown in FIGS. 1-5 of the description, 18 hybridoma cell strains to produce antibodies that specifically bind to hCLDN18.2 were finally obtained.

(74) The hybridoma cell strain secreting the monoclonal antibody obtained by the screening was cultured, and the total RNA was extracted from the cell by a conventional biological method. cDNA was synthesized from the total RNA template via reverse transcription, using PrimeScript™ 1st Strand cDNA Synthesis Kit (TAKARA).

(75) The cDNA then served as a template in the amplification using antibody constant region

primers. After the PCR products were separated by agarose gel electrophoresis, the DNA fragments were purified and recovered, and the amino acid sequences of the variable regions of 18 monoclonal antibodies of the present disclosure were obtained by sequencing, and the results were shown in table 1:

(76) TABLE-US-00001 TABLE 1 Amino acid sequences of the variable regions of 18 monoclonal antibodies Clone Variable region of Variable region of Number heavy chain (VH) light chain (VL)
59F9 SEQ ID NO: 1 SEQ ID NO: 2 45H1 SEQ ID NO: 3 SEQ ID NO: 4 43F5 SEQ ID NO: 5 SEQ ID NO: 6 4007 SEQ ID NO: 7 SEQ ID NO: 8 76D11 SEQ ID NO: 9 SEQ ID NO: 10 32G8 SEQ ID NO: 11 SEQ ID NO: 12 37G8 SEQ ID NO: 13 SEQ ID NO: 14 39C8 SEQ ID NO: 15 SEQ ID NO: 16 4212 SEQ ID NO: 17 SEQ ID NO: 18 126A1B SEQ ID NO: 19 SEQ ID NO: 20 239E9 SEQ ID NO: 21 SEQ ID NO: 22 22909 SEQ ID NO: 23 SEQ ID NO: 24 252F11 SEQ ID NO: 25 SEQ ID NO: 26 299B2 SEQ ID NO: 27 SEQ ID NO: 28 24117 SEQ ID NO: 29 SEQ ID NO: 30 251H1B SEQ ID NO: 31 SEQ ID NO: 32 287F12 SEQ ID NO: 33 SEQ ID NO: 34 25304 SEQ ID NO: 35 SEQ ID NO: 36

(77) Based on the above amino acid sequences, the CDRs and FRs of the variable regions of the antibodies were divided using Kabat numbering rules, and the composition of the 6 CDR sequences of each antibody was shown in Table 2 below, wherein the numbers in parentheses in Table 2 indicate the sequence numbers, e.g., (37) represents SEQ ID NO: 37.

(78) TABLE-US-00002 TABLE 2 CDRs of 18 monoclonal antibodies Clone Number
HCDR1 HCDR2 HCDR3 LCDR1 LCDR2 LCDR3 59F9 GYWIE EILLGSGSIKYNVKFKD
KGLRGNSFDY KSSQSLLNSGNQKSYLT WASTRES QNDYYYPFT (SEQ ID (SEQ ID NO: 38)(SEQ ID (SEQ ID NO: 86)(SEQ ID (SEQ ID NO: 97) NO: 37) NO: 39) NO: 93) 48H1 NYWTH MIHPNSGSSNYNEKFKS IHYGNSMDY
KSSQSLLFNSGNQKKNYLT WASTWES QNAYSYPFT (SEQ ID (SEQ ID NO: 41) (SEQ ID (SEQ ID NO: 87)(SEQ ID (SEQ ID NO: 98) NO: 40) NO: 42) NO: 94) 43F5 SYWTH MIHPNSGSSNYNEKFKS IHYGNAMDY KSSQSLLNSGNQKKNYLT
WASTRES QNDYSFPFT (SEQ ID (SEQ ID NO: 41)(SEQ ID (SEQ ID NO: 88)(SEQ ID (SEQ ID NO: 99) NO: 43) NO: 44) NO: 93) 40G7 NYWMH
MIHPNSYSTNYNEKFRS IYYGNAMDY KSSQSLLFNSGNQKKNYLT WASTRES QNDYSYPFT (SEQ ID (SEQ ID NO: 46)(SEQ ID (SEQ ID NO: 87)(SEQ ID (SEQ ID NO: 100) NO: 45) NO: 47) NO: 95) 76D11 GYWIE EILPGSGSIKYNEKFKD
KGLRGNSFDY KSSQSLLNSGNQKKNYLT WASTRES QNDYYYPFT (SEQ ID (SEQ ID NO: 48)(SEQ ID (SEQ ID NO: 88)(SEQ ID (SEQ ID NO: 97) NO: 37) NO: 39) NO: 93) 32G8 DYHMN VINPYNGGIRYNQKFKG IYYGNSFAY
KSSQSLLNSGNQKKNYLT WASTRES QNNYIYPFT (SEQ ID (SEQ ID NO: 50) (SEQ ID (SEQ ID NO: 88)(SEQ ID (SEQ ID NO: 101) NO: 49) NO: 51) NO: 93) 37G8 DYHMN LINPYNGGIRFNQKFKG IYYGNSFAY KSGQSLLNSGNQKKNYLT
WASTRES QNDYFYPYT (SEQ ID (SEQ ID NO: 52)(SEQ ID (SEQ ID NO: 89)(SEQ ID (SEQ ID NO: 102) NO: 49) NO: 51) NO: 93) 39C8 SYWMI
QIYPGDGDTNYNGKFK IYYGNAFAY KSSQSLLNSGNQKKNYLT WASTRES QNDYSYPFT (SEQ ID G (SEQ ID (SEQ ID NO: 88)(SEQ ID (SEQ ID NO: 100) NO: 53)(SEQ ID NO: 54) NO: 55) NO: 93) 42D2 DYQMN FINPYNGGIRYNQKFKG
IYFGNSFAN KPSQSLLNSGNQKKNYLT WASTRES QNDYIYPYT (SEQ ID (SEQ ID NO: 57)(SEQ ID (SEQ ID NO: 90)(SEQ ID (SEQ ID NO: 103) NO: 56) NO: 58) NO: 93) 126A11 TYGVS VIWGDGSTNYHSALIS PGLRNAMDY
KSSQSLLNSGNQKKNYLA GASTRES QNDLIYPLT (SEQ ID (SEQ ID NO: 60) (SEQ ID (SEQ ID NO: 91)(SEQ ID (SEQ ID NO: 104) NO: 59) NO: 61) NO: 96) 239E9 DYTMMH FIGVYYGNTNYNQKFKG IGRGNAMDY
KSSQSLLNSGNQKKNYLT WASTRES QNAYSYPFT (SEQ ID (SEQ ID NO: 63) (SEQ ID (SEQ ID NO: 88)(SEQ ID (SEQ ID NO: 98) NO: 62) NO: 64)

NO: 93) 229C9 SGYSWH YIHYSGGTNYNPSLKS LERGNSFAY KSTQSLNLSGNQKNYLT
WASTRES QNDYFYFPT (SEQ ID (SEQ ID NO: 66) (SEQ ID (SEQ ID NO:
92) (SEQ ID SEQ ID NO: 105) NO: 65) NO: 67) NO: 93) 252F11 INYVMS
BIRTGGDYTTYVDTVTG VGYGNSLDY KSSQSLLNSGNQKNYLT WASTRES QNNYFYPLT
(SEQ ID (SEQ ID NO: 69) (SEQ ID (SEQ ID NO: 88) (SEQ ID (SEQ
ID NO: 106) NO: 68) NO: 70) NO: 93) 299B2 NYWIH RIYPGTGNTYYNEKFTG
EGYGKGNMDY KSSQSLLNSGNQKNYLT WASTRES QNAYYYPYT (SEQ ID (SEQ
ID NO: 72) (SEQ ID (SEQ ID NO: 88) (SEQ ID (SEQ ID NO: 107) NO:
71) NO: 73) NO: 93) 241F7 AYNMN NIDPYYGGTNYNQKFKG VYYGNSLIY
KSSQSLLNSGNQKNYLT WASTRES QNNYFYPLT (SEQ ID (SEQ ID NO: 75)
(SEQ ID (SEQ ID NO: 88) (SEQ ID (SEQ ID NO: 106) NO: 74) NO: 76)
NO: 93) 251H11 TFWIH KIYPGTGYTYNEKFKG EGYGKGNAVDF
KSSQSLLNSGNQKNYLT WASTRES QNDYTYPST (SEQ ID (SEQ ID NO: 78)
(SEQ ID (SEQ ID NO: 87) (SEQ ID (SEQ ID NO: 108) NO: 77) NO: 79)
NO: 93) 287F12 TAGMH WTNTHSGEPKYAEDFKG WGRGNALDY
KSSQSLLNSGNQKNYLT WASTRES QNTYSYPLT (SEQ ID (SEQ ID NO: 81)
(SEQ ID SEQ ID NO: 88) (SEQ ID (SEQ ID NO: 109) NO: 80) NO: 82)
NO: 93) 253 C4 SYWIH RFYPGTGTAYYNENFEG EGYGKGNAMDY
KSSQSLLNSGNQKNYLT WASTRES QNDYFYFPT (SEQ ID (SEQ ID NO: 84)
(SEQ ID (SEQ ID NO: 88) (SEQ ID SEQ ID NO: 110) NO: 83) NO: 85)
NO: 93)

Example 3 Construction of an Anti-CLDN18.2 Chimeric Antibody and Transient Transfection Expression of the Same in Eukaryotic Cells

(79) The target gene fragment generated after splicing the sequenced monoclonal antibody variable region of the present disclosure and the human IgG1 constant region was cloned into a pcDNA3.4 expression vector to prepare a transfection-grade expression plasmid. Expi293F™ cells (Thermo Fisher Scientific) were cultured in a serum-free medium, seeded in shake flasks (Coming Inc.), and cultured on a shaking table in an environment of 37° C., 8% CO.sub.2. The cell density was adjusted, the recombinant expression vector containing the target gene fragment and the ExpiFectamine™ 293 transfection reagent were mixed according to an appropriate ratio and added into a cell culture shake flask, after transfection 16-18 h, ExpiFectamine™ 293 Transfection Enhancer 1 and ExpiFectamine™ 293 Transfection Enhancer 2 were added, the supernatant was collected and purified after 6 days of cell culture, and finally purified chimeric antibody was subjected to SDS-PAGE purity analysis and A280 concentration determination. Chimeric antibodies were named in such a way that the prefix ch- was added based on the original hybridoma clone.

Example 4 Binding Assay of Anti-CLDN18.2 Chimeric Antibody

(80) a. Binding of Anti-CLDN18.2 Chimeric Antibodies to Cells Expressing hCLDN18.2

(81) FACS was used to detect the binding of anti-CLDN18.2 chimeric antibodies to HEK293 cells stably transfected expressing hCLDN18.2 (HEK293-hCLDN18.2) and gastric cancer tumor tissue-derived cells naturally expressing hCLDN18.2 (PDX-hCLDN18.2).

(82) HEK293-hCLDN18.2 or PDX-hCLDN18.2 cells were harvested and resuspended in PBS to adjust cell concentration, and the gradiently diluted antibody was added, with irrelevant human IgG being a negative control and chimeric antibody ch-175D10 from patent CN103509110B being a positive control (reference antibody). Following incubation in a 4° C. shaking table for 50 min-1 h, the mixture was centrifugally washed twice with phosphate buffer solution, added with fluorescently labeled anti-human IgG secondary antibody, 100 µL per well; after incubation in a 4° C. shaking table for 40 min-1 h, the mixture was centrifugally washed twice with phosphate buffer solution, and then the prepared sample was detected on a flow cytometer; the mean fluorescence intensity (hereinafter referred to as MFI) for each concentration was calculated by the software, and

then the half binding concentration (hereinafter referred to as EC.sub.50) and the mean maximum fluorescence intensity (Top MFI) were calculated by GraphPad software, and the results were shown in Table 3.

(83) TABLE-US-00003 TABLE 3 Binding of anti-CLDN18.2 chimeric antibodies to hCLDN18.2 HEK293-hCLDN18.2 PDX-hCLDN18.2 Mean maximum fluorescence EC.sub.50 intensity EC.sub.50 intensity Number (nM) (Top MFI) (nM) (Top MFI) ch-175D10 0.881 5451 63.85 7160 ch-59F9 2.411 9038 12.38 18084 ch-48H1 2.922 9485 5.588 13614 ch-43F5 1.510 5586 5.354 7616 ch-40G 2.309 7467 7.687 10118 ch-76D11 2.464 8718 12.31 16836 ch-32G8 2.657 8645 5.392 14764 ch-37G8 5.048 10376 6.999 14399 ch-39C8 2.700 7820 20.14 17548 ch-42D2 2.416 10385 7.418 15709 ch-126A11 2.121 9972 14.99 17121 ch-239E9 2.308 9269 4.554 14552 ch-229C9 2.457 9869 10.67 15716 ch-252F11 2.202 8383 15.82 16505 ch-299B2 1.524 10126 5.809 16342 ch-241F7 3.024 12203 4.871 15220 ch-251H11 2.721 10790 96.23 19876 ch-287F12 2.650 10947 16.78 17591 ch-253C4 2.275 10345 9.735 15546

(84) Table 3 and FIGS. 6-17 showed the affinity results for the chimeric antibody of the present disclosure and the reference antibody ch-175D10 to HEK293-hCLDN18.2 cells and PDX-hCLDN18.2 cells, respectively. Experimental results showed that: the binding of the chimeric antibody of the present disclosure to HEK293-hCLDN18.2 cells exhibited a mean maximum fluorescence intensity of 5586-12203, whereas the binding of the reference antibody ch-175D10 to HEK293-hCLDN18.2 under the same reaction conditions exhibited a mean maximum fluorescence intensity of only 5451. The binding of the chimeric antibody to PDX-hCLDN18.2 exhibited a mean maximum fluorescence intensity of 7616-19876, a half binding concentration (EC.sub.50) of 4.554-96.23 nM, whereas the binding of the reference antibody ch-175D10 to PDX-hCLDN18.2 under the same reaction conditions exhibited a mean maximum fluorescence intensity of only 7160 and a half binding concentration (EC.sub.50) of only 63.85 nM. Thus, the binding of most chimeric antibodies of the present disclosure to hCLDN18.2 antigen was more strongly than that of ch-175D10.

(85) B. Binding Selectivity of Anti-CLDN18.2 Chimeric Antibodies

(86) FACS was used to detect the binding of chimeric antibodies of the present disclosure to HEK293 cells stably transfected expressing murine CLDN18.2 (HEK293-mCLDN18.2) and HEK293 cells stably transfected expressing human CLDN18.1 (HEK293-hCLDN18.1).

(87) HEK293-mCLDN18.2 and HEK293-hCLDN18.1 cells were harvested separately and resuspended in PBS to adjust cell concentration, and the gradiently diluted chimeric antibody was added, wherein irrelevant human IgG was used as a negative control and ch-175D10 was still a positive control (reference antibody). Following incubation in a 4° C. shaking table for 50 min, the mixture was centrifugally washed twice with phosphate buffer solution, added with fluorescently labeled anti-human IgG secondary antibody, 100 µL per well; after incubation in a 4° C. shaking table for 40 min, the mixture was centrifugally washed twice with phosphate buffer solution, and then the prepared sample was detected on a flow cytometer; the half binding concentration (EC.sub.50) and the mean maximum fluorescence intensity (Top MFI) were calculated by GraphPad software, and the results were shown in Table 4.

(88) TABLE-US-00004 TABLE 4 Binding of anti-CLDN18.2 chimeric antibodies to mCLDN18.2 and hCLDN18.1, respectively HEK293-mCLDN18.2 Mean maximum fluorescence EC.sub.50 intensity HEK293-hCLDN18.1 Clone Number (nM) (Top MFI) Binding or not (+/-) ch-175D10 1.014 4187 - ch-59F9 1.222 11162 - ch-48H1 1.763 14768 - ch-43F5 1.132 6093 - ch-40G7 1.115 10037 - ch-76D11 0.702 6337 - ch-32G8 0.840 6870 - ch-37G8 1.712 8926 - ch-39C8 1.627 10599 - ch-42D2 0.764 7400 - ch-126A11 0.573 7636 - ch-239E9 1.241 12726 - ch-229C9 1.188 12740 - ch-252F11 0.948 8975 - ch-299B2 0.758 13595 - ch-241F7 1.160 10495 - ch-251H11 1.009 8588 - ch-287F12 1.138 10191 - ch-253C4 0.790 8326 -

(89) Table 4 showed the affinity results for the chimeric antibody of the present disclosure and the reference antibody ch-175D10 to HEK293-mCLDN18.2 cells and HEK293-hCLDN18.1 cells,

respectively. Experimental results showed that: the chimeric antibody of the present disclosure was the same as the reference antibody ch-175D10, both of which bound to the mCLDN18.2 antigen, wherein the binding of the chimeric antibody to HEK293-mCLDN18.2 exhibited a mean maximum fluorescence intensity of 6093-14768 and a half binding concentration (EC.sub.50) of 0.573-1.763 nM, the binding of the reference antibody ch-175D10 to HEK293-mCLDN18.2 under the same reaction conditions exhibited a mean maximum fluorescence intensity of 4187 and a half binding concentration (EC.sub.50) of 1.014 nM, indicating that the chimeric antibody exhibited the binding EC.sub.50 comparable to that of the reference antibody, and the maximum binding higher than that of the reference antibody. Moreover, the chimeric antibody was the same as the reference antibody ch-175D10, neither binding to the hCLDN18.1 antigen.

Example 5 In Vitro Functional Assay of Anti-CLDN18.2 Chimeric Antibodies

(90) a. Antibody-Dependent Cell-Mediated Cytotoxicity (ADCC)

(91) CHO-K1/hCLDN18.2 cells were used as target cells, NK cells transfected with 158V/V type FcγRIIIa gene (NK92/FcRγ3a.158V/V) were used as effector cells, and the release of lactate dehydrogenase (LDH) in cell was detected by the cytotoxicity assay kit (Roche) and used as an indicator of cell killing effect.

(92) CHO-K1/hCLDN18.2 cells were harvested by centrifugation, the supernatant was discarded, and the cells were resuspended in ADCC buffer to adjust the cell density and transferred to a 96-well assay plate. Chimeric antibodies, control sample working solutions, or ADCC buffer at different concentration gradients were transferred to a 96-well plate, incubated for about 30 min in a cell incubator (37° C./5% CO.sub.2), effector cells, ADCC buffer, or cell lysates were transferred to a 96-well assay plate, and incubated for an additional about 6 h in a cell incubator (37° C./5% CO.sub.2). After the incubation, the 96-well assay plate was removed and centrifuged at a low rotating speed, the supernatant was pipetted carefully and transferred into a new 96-well assay plate, LDH detection working solution was added, the plate was incubated at room temperature for about 10-30 min, the OD value was detected on a microplate reader, wherein a detection wavelength was 492 nm, and a reference wavelength was 650 nm.

(93) The percentage of cell lysis caused by the ADCC effect was calculated using the following formula: % cell lysis=100%× (sample release-target cell/effector cell mixed release)/(maximum release-target cell release), wherein the maximum release was the absorbance value produced in the wells of target cells treated with Triton X-100, the target cell/effector cell mixed release was the absorbance value produced in the wells of target cells and effector cell mixture, and the target cell release was the absorbance value produced in the wells containing only target cells, the sample release was the absorbance values produced in the wells of chimeric antibody, target cells, and effector cells mixture, and EC.sub.50 and maximal lysis were calculated by GraphPad software, and the results were shown in Table 5.

(94) TABLE-US-00005 TABLE 5 ADCC activity of anti-CLDN18.2 chimeric antibodies

Clone Number	Maximum lysis (%)	EC.sub.50 (μg/ml)
ch-175D10	31.81	0.021
ch-59F9	29.37	0.023
ch-48H1	44.73	0.011
ch-43F5	48.46	0.064
ch-40G7	32.72	0.018
ch-76D11	45.60	0.021
ch-32G8	46.83	0.019
ch-37G8	31.71	0.022
ch-39C8	34.45	0.022
ch-42D2	39.12	0.020
ch-126A11	36.65	0.010
ch-239E9	36.68	0.021
ch-229C9	29.06	0.023
ch-252F11	31.09	0.022
ch-299B2	33.52	0.047
ch-241F7	36.57	0.027
ch-251H11	38.79	0.024
ch-287F12	35.46	0.017
ch-253C4	35.72	0.036

(95) Table 5 and FIGS. 18-23 showed ADCC results for the chimeric antibody of the present disclosure and the reference antibody ch-175D10 on CHO-K1/hCLDN18.2 cells. Experimental results showed that: the maximal ADCC effect of the chimeric antibody of the present disclosure on CHO-K1/hCLDN18.2 cells was 29.06%-48.46%, and the ADCC effect of the reference antibody ch-175D10 was 31.81% under the same reaction conditions. The concentration of the chimeric antibody of the present disclosure that produces 50% ADCC effect (EC.sub.50) was 0.010-0.064 μg/mL, and the concentration of ch-175D10 that produces 50% ADCC effect (EC.sub.50) under the same reaction conditions was 0.021 μg/mL. The above results demonstrated

that the chimeric antibodies of the present disclosure were comparable to the reference antibody ch-175D10 in ADCC activity.

(96) B. Complement-Dependent Cytotoxicity (CDC)

(97) Cell viability was measured by CellTiter-Glo® chemiluminescent cell viability assay kit (Promega) using CHO-K1/hCLDN18.2 as target cells and pooled normal human serum (PNHS) as complement source.

(98) CHO-K1/hCLDN18.2 cells were harvested, resuspended in CDC buffer to adjust cell density, seeded into a 384-well cell plate, 20 µL per well, to prepare a 4-fold concentration of sample solution, which (CDC buffer as control) was transferred to the corresponding wells of the 384-well cell plate, 10 µL per well, the plate was incubated at room temperature for about 30 min, and the pooled normal human serum (PNHS) was diluted to the 4-fold working concentration using CDC buffer. The diluted PNHS was transferred to the corresponding wells of the incubated 384-well plate, 10 µL per well, the plate was incubated in a cell incubator (37° C./5% CO₂) for about 4 h, the 384-well plate was removed and assayed with CellTiter-Glo® chemiluminescent cell viability assay kit (Promega), and the results were read using PHERAstar Plus software.

(99) The cell lysis rate caused by the chimeric antibody in the CDC assay was calculated using the following formula:

$$\% \text{ cell lysis} = 100\% \times (1 - (\text{test wells} - \text{serum control wells}) / (\text{cell} + \text{serum wells} - \text{serum control wells}))$$

(100) The experimental controls are: the serum control wells: only serum (i.e., 30 µL buffer+10 µL diluted serum). Cells+serum wells: serum was added to wells of CHO-K1/hCLDN18.2 cell suspension (i.e., 20 µL cell suspension+10 µL buffer+10 µL diluted serum). Test wells: serum and chimeric antibody were added to wells of CHO-K1/hCLDN18.2 cell suspension (i.e., 20 µL cell suspension+10 µL antibody+10 µL diluted serum). The EC₅₀ and maximum lysis were calculated using GraphPad software, and the results were shown in Table 6.

(101) TABLE-US-00006 TABLE 6 CDC activity of anti-CLDN18.2 chimeric antibodies

Clone Number	Maximum lysis (%)	EC ₅₀ (µg/ml)
ch-175D10	87.41	1.193
ch-59F9	94.71	0.091
ch-48H1	94.27	0.170
ch-43F5	89.79	0.206
ch-40G7	89.08	0.380
ch-76D11	95.36	0.108
ch-32G8	93.85	0.108
ch-37G8	92.86	0.160
ch-39C8	94.75	0.177
ch-42D2	94.47	0.130
ch-126A11	95.02	0.065
ch-239E9	94.80	0.139
ch-229C9	96.11	0.165
ch-252F11	95.44	0.179
ch-299B2	95.53	0.124
ch-241F7	95.05	0.099
ch-251H11	97.11	0.359
ch-287F12	96.32	0.141
ch-253C4	95.23	0.038

(102) Table 6 and FIGS. 24-29 showed CDC results for the chimeric antibody of the present disclosure and the reference antibody ch-175D10 on CHO-K1/hCLDN18.2 cells. Experimental results showed that: the maximal CDC effect of the chimeric antibody of the present disclosure on CHO-K1/hCLDN18.2 cells was 89.08%-97.11%, and the maximal CDC effect of the reference antibody ch-175D10 was 87.41% under the same reaction conditions. The concentration of the chimeric antibody of the present disclosure that produces 50% CDC effect (EC₅₀) was 0.038-0.380 pg/mL, and the concentration of ch-175D10 that produces 50% CDC effect (EC₅₀) under the same reaction conditions was higher than 1 µg/mL. It can be seen that the CDC activity of most antibodies was stronger than that of the reference antibody.

Example 6 Preparation of Variants of Anti-CLDN18.2 Chimeric Antibodies

(103) Through post-translational modification (PTM) analysis of the monoclonal antibody disclosed by the present disclosure, it was found that 1 deamidation site was present in both variable regions of 299B2 and 253C4; single site-directed mutagenesis was performed on the 31st, or 32nd, or 33rd amino acid in the variable region of the light chain of 299B2 to prepare three mutants of 299B2: 299B2-N31Q, 299B2-S32A, and 299B2-G33A, respectively.

(104) And single site-directed mutagenesis was performed on the 31st, or 32nd, or 33rd amino acid in the variable region of the light chain of 253C4 to prepare three mutants of 253C4: 253C4-N31Q, 253C4-S32A, and 253C4-G33A, respectively.

(105) The amino acid sequences of the heavy chain variable region of 299B2-N31Q, 299B2-S32A, and 299B2-G33A were the same as that of 299B2, the amino acid sequences of the heavy chain

variable region of 253C4-N31Q, 253C4-S32A, and 253C4-G33A were the same as that of 253C4. The variable region CDR sequences and heavy and light chain variable region sequences of the 6 variants described above were shown in Tables 7 and 8, respectively.

(106) TABLE-US-00007 TABLE 7 CDR sequences of antibodies 299B2 and 253C4 and variants thereof Clone Number HCDR1 HCDR2 HCDR3 LCDR1 LCDR2 LCDR3 299B2 NYWIIH RIYPGTGNTYYNEKFTG EGYGKGNSMDY KSSQSLLNSGNQKNYLT WASTRES QNAYYYPYT (SEQ ID NO: 72) (SEQ ID NO: 88) (SEQ ID NO: 71) (SEQ ID NO: 73) (SEQ ID NO: 93) (SEQ ID NO: 107) 299B2- NYWIIH RIYPGTGNTYYNEKFTG EGYGKGNSMDY KSSQSLLQSGNQKNYLT WASTRES QNAYYYPYT N31Q (SEQ ID NO: 72) (SEQ ID NO: 111) (SEQ ID NO: 71) (SEQ ID NO: 73) (SEQ ID NO: 93) (SEQ ID NO: 107) 299B2- NYWIIH RIYPGTGNTYYNEKFTG EGYGKGNSMDY KSSQSLLNAGNQKNYLT WASTRES QNAYYYPYT S32A (SEQ ID NO: 72) (SEQ ID NO: 112) (SEQ ID NO: 71) (SEQ ID NO: 73) (SEQ ID NO: 93) (SEQ ID NO: 107) 299B2- NYWIIH RIYPGTGNTYYNEKFTG EGYGKGNSMDY KSSQSLLNSANQKNYLT WASTRES QNAYYYPYT G33A (SEQ ID NO: 72) (SEQ ID NO: 113) (SEQ ID NO: 71) (SEQ ID NO: 73) (SEQ ID NO: 93) (SEQ ID NO: 107) 253C4 SYWIIH RFYPGTGTAYYNENFEG EGYGKGNSMDY KSSQSLLNSGNQKNYLT WASTRES QNDYYFPFT (SEQ ID NO: 84) (SEQ ID NO: 88) (SEQ ID NO: 83) (SEQ ID NO: 85) (SEQ ID NO: 93) (SEQ ID NO: 110) 253C4- SYWIIH RFYPGTGTAYYNENFEG EGYGKGNSMDY KSSQSLLQSGNQKNYLT WASTRES QNDYYFPFT N31Q (SEQ ID NO: 84) (SEQ ID NO: 111) (SEQ ID NO: 83) (SEQ ID NO: 85) (SEQ ID NO: 93) (SEQ ID NO: 110) 253C4- SYWIIH RFYPGTGTAYYNENFEG EGYGKGNSMDY KSSQSLLNAGNQKNYLT WASTRES QNDYYFPFT S32A (SEQ ID NO: 84) (SEQ ID NO: 112) (SEQ ID NO: 83) (SEQ ID NO: 85) (SEQ ID NO: 93) (SEQ ID NO: 110) 253C4- SYWIIH RFYPGTGTAYYNENFEG EGYGKGNSMDY KSSQSLLNSANQKNYLT WASTRES QNDYYFPFT G33A (SEQ ID NO: 84) (SEQ ID NO: 113) (SEQ ID NO: 83) (SEQ ID NO: 85) (SEQ ID NO: 93) (SEQ ID NO: 110)

(107) TABLE-US-00008 TABLE 8 Variable region sequences of antibodies 299B2 and 253C4 and variants thereof Clone Number Heavy chain variable region Light chain variable region 299B2 SEQ ID NO: 27 SEQ ID NO: 28 299B2-N31Q SEQ ID NO: 27 SEQ ID NO: 114 299B2-S32A SEQ ID NO: 27 SEQ ID NO: 115 299B2-G33A SEQ ID NO: 27 SEQ ID NO: 116 253C4 SEQ ID NO: 35 SEQ ID NO: 36 253C4-N31Q SEQ ID NO: 35 SEQ ID NO: 117 253C4-S32A SEQ ID NO: 35 SEQ ID NO: 118 253C4-G33A SEQ ID NO: 35 SEQ ID NO: 119

(108) As described in Example 3, a chimeric antibody was constructed by subcloning the target gene fragment produced by splicing the 6 variant variable regions described above with the human IgG1 constant region into an expression vector, and the chimeric antibody of the variant was named in such a way that the prefix ch- was added based on the “hybridoma clone-mutation site”, for example, a chimeric antibody of 299B2-N31Q was designated as ch-299B2-N31Q. Following transient expression in mammalian cell lines, affinity assays were performed using HEK293-hCLDN18.2 cells and monoclonal cell supernatants were screened by PACS. Affinity results for variants of antibodies ch-299B2 and ch-253C4 are shown in Table 9.

(109) TABLE-US-00009 TABLE 9 Affinity of variants of antibodies ch-299B2 and ch-253C4 EC.sub.50 Mean maximum fluorescence Clone Number (ng/ml) intensity (Top MFI) ch-175D10 155.9 2836 ch-299B2-X31Q —* —* ch-299B2-S32A 1307 6504 ch-299B2-G33A 157.0 6869 ch-253C4-N31Q 175.0 6565 ch-253C4-S32A 206.7 6894 ch-253C4-G33A 186.7 6837 *The maximum concentration has not reached the maximum binding, and the curve cannot be fitted.

Example 7 Preparation of Humanized Antibodies

(110) A. Humanized Design and Expression of Antibody 299B2-S32A

(111) Through sequence similarity comparison, the antibody germline with the highest similarity to 299B2 was selected as an antibody template. In this example, the IMGT database IGHV1-46*01 was selected as an antibody template for the heavy chain of 299B2-S32A, IGKV4-1*01 was selected as an antibody template for the light chain of 299B2-S32A, the CDR regions of the antibody template were replaced with the CDR regions of the light chain and heavy chain of 299B2-S32A.

(112) Homologous modeling of murine antibody variable region sequences was performed. The best modeling template was searched in the PDB antibody database based on the sequence of the murine antibody variable region, and 2GKI with 74% homology was selected as the template. Based on the spatial structure of 2GKI, the amino acid residues in the framework region of the CDR-grafted sequence were back-mutated according to the following criteria: 1. the classical residues in the framework region was selected for back mutation; 2. the residues of the hydrophobic core region in the framework region was selected for back mutation; 3. the residues in the heavy chain/light chain interaction interface was selected for back mutation; 4. similar residues was also selected for low priority back mutations.

(113) 299B2-S32A was humanized to obtain 4 humanized antibodies hu299B2-S32A-1, hu299B2-S32A-2, hu299B2-S32A-3, and hu299B2-S32A-4, the sequences of all humanized antibodies of 299B2-S32A mentioned above were shown in Table 10.

(114) The target gene fragment which was generated by splicing the humanized antibody hu299B2-S32A variable region and the human IgG1 constant region was subcloned into a pcDNA3.4 expression vector via standard methods known to those skilled in the art, Expi293F™ cells in a logarithmic growth phase were transiently transfected by an ExpiFectamine™ 293 transfection reagent, and culture supernatant was collected and subjected to affinity purification, the final purified antibody was subjected to SDS-PAGE purity analysis and A280 concentration determination.

(115) B. Humanized Design and Expression of Antibody 253C4-N31Q

(116) Through sequence similarity comparison, the antibody germline with the highest similarity to 253C4 was selected as an antibody template. In this example, the IMGT database IGKV4-1*01 was selected as an antibody template for the heavy chain of 253C4-N31Q, IGHV1-2*06 was selected as an antibody template for the light chain of 253C4-N31Q, the CDR regions of the antibody template were replaced with the CDR regions of the light chain and heavy chain of 253C4-N31Q.

(117) Homologous modeling of murine antibody variable region sequences was performed. The best modeling template was searched in the PDB antibody database based on the sequence of the murine antibody variable region, and 2GKI with 74% homology was selected as the template. Based on the spatial structure of 2GKI, the amino acid residues in the framework region of the CDR-grafted sequence were back-mutated according to the following criteria: 1. the classical residues in the framework region was selected for back mutation; 2. the residues of the hydrophobic core region in the framework region was selected for back mutation; 3. the residues in the heavy chain/light chain interaction interface was selected for back mutation; 4. similar residues was also selected for low priority back mutations.

(118) 253C4-N31Q was humanized to obtain 3 humanized antibodies hu253C4-N31Q-1, hu253C4-N31Q-2, and hu253C4-N31Q-3, the sequences of all 253C4-N31Q humanized antibodies were shown in Table 10.

(119) The target gene fragment which was generated by splicing the humanized antibody hu253C4-N31Q variable region and the human IgG1 constant region was subcloned into a pcDNA3.4 expression vector via standard methods known to those skilled in the art, Expi293F™ cells in a logarithmic growth phase were transiently transfected by an ExpiFectamine™ 293 transfection reagent, and culture supernatant was collected and subjected to affinity purification, the final purified antibody was subjected to SDS-PAGE purity analysis and A280 concentration determination.

(120) TABLE-US-00010 TABLE 10 Variable region sequences of humanized antibodies hu299B2-S32A and hu253C4-N31Q Heavy chain variable Light chain variable Clone Number region region hu299B2-S32A-1 SEQ ID NO: 120 SEQ ID NO: 121 hu299B2-S32A-2 SEQ ID NO: 120 SEQ ID NO: 123 hu299B2-S32A-3 SLQ ID NO: 120 SEQ ID NO: 124 hu299B2-S32A-4 SEQ ID NO: 122 SEQ ID NO: 121 hu253C4-N31Q-1 SEQ ID NO: 125 SEQ ID NO: 126 hu253C4-N31Q-2 SEQ ID NO: 125 SEQ ID NO: 127 hu253C4-N31Q-3 SEQ ID NO: 128 SEQ ID NO: 126

Example 8 Binding Activity of Humanized Antibodies Hu299B2 and Hu253C4

(121) A. Binding of Humanized Antibodies to Cells Expressing hCLDN18.2

(122) Binding activity assays were performed using HEK293-hCLDN18.2 and PDX-hCLDN18.2 cells, with reference to Example 4A, with irrelevant human IgG being a negative control and chimeric antibody ch-175D10 from patent CN103509110B being a positive control (reference antibody).

(123) Using flow cytometry measurement, the mean fluorescence intensity (hereinafter referred to as MFI) for each concentration was calculated by the software, and then the half binding concentration (EC.sub.50) and the mean maximum fluorescence intensity (Top MFI) were calculated by GraphPad software, and the results were shown in Table 11.

(124) TABLE-US-00011 TABLE 11 Binding of anti-CLDN18.2 humanized antibody to hCLDN18.2 HEK293-hCLDN18.2 PDX-hCLDN18.2 Mean maximum Mean maximum fluorescence fluorescence EC.sub.50 intensity EC.sub.50 intensity Clone Number (nM) (Top MEI) (nM) (Top MFI) ch-175D10 1.434 26921 24.89 5703 hu299B2-S32A-1 0.407 29529 29.68 25189 hu299B2-S32A-2 0.463 36728 24.60 22877 hu299B2-S32A-3 0.399 39986 26.12 24291 hu299B2-S32A-4 0.318 31234 29.26 22967 hu253C4-N31Q-1 0.216 23017 12.02 16291 hu253C4-N31Q-2 0.361 29037 11.42 15585 hu253C4-N31Q-3 0.145 18772 25.57 21616

(125) Table 11 and FIGS. 30-33 showed the affinity results for the humanized antibody of the present disclosure and the reference antibody ch-175D10 to HEK293-hCLDN18.2 cells and PDX-hCLDN18.2 cells, respectively. Experimental results showed that: the binding of hu299B2-S32A humanized antibody of the present disclosure to HEK293-hCLDN18.2 cells exhibited a mean maximum fluorescence intensity of 29529-39986, whereas the binding of the reference antibody ch-175D10 to HEK293-hCLDN18.2 under the same reaction conditions exhibited a mean maximum fluorescence intensity of only 26921, indicating that the affinity of the humanized antibody hu299B2-S32A to HEK293-hCLDN18.2 was superior to that of the reference antibody ch-175D10. The binding of hu253C4-N31Q to HEK293-hCLDN18.2 exhibited the mean maximum fluorescence intensity comparable to that of the reference antibody ch-175D10, and EC.sub.50 was superior to that of the reference antibody ch-175D10.

(126) The binding of hu299B2-S32A and hu253C4-N31Q of the present disclosure to PDX-hCLDN18.2 exhibited a mean maximum fluorescence intensity of 15585-25189, a half binding concentration (EC.sub.50) of 11.42-29.68 nM, whereas the binding of the reference antibody ch-175D10 to PDX-hCLDN18.2 under the same reaction conditions exhibited a mean maximum fluorescence intensity of only 5703, and a half binding concentration (EC.sub.50) 24.89 nM. It can be seen that the maximum binding of the humanized antibody of the present disclosure to the native hCLDN18.2 antigen was stronger than that of ch-175D10, and EC.sub.50 was comparable to that of ch-175D10.

(127) B. Binding Selectivity of Humanized Antibodies

(128) The binding of the humanized antibody of the present disclosure to HEK293-mCLDN18.2 cells and HEK293-hCLDN18.1 cells was examined using FACS according to the method described in Example 4B.

(129) HEK293-mCLDN18.2 and HEK293-hCLDN18.1 cells were harvested separately and resuspended in PBS to adjust cell concentration, and gradiently diluted humanized antibody was added, wherein irrelevant human IgG was used as a negative control and ch-175D10 was still a positive control (reference antibody). Following incubation in a 4° C. shaking table for 50 min, the

mixture was centrifugally washed twice with phosphate buffer solution, added with fluorescently labeled anti-human IgG secondary antibody, 100 μ L per well; after incubation in a 4° C. shaking table for 40 min, the mixture was centrifugally washed twice with phosphate buffer solution, and then the prepared sample was detected on a flow cytometer; the half binding concentration (EC.sub.50) and the mean maximum fluorescence intensity (Top MFI) were calculated by GraphPad software, and the results were shown in Table 12.

(130) TABLE-US-00012 TABLE 12 Binding of anti-CLDN18.2 humanized antibodies to mCLDN18.2 and hCLDN18.1, respectively HEK293-mCLDN18.2 Mean maximum fluorescence EC.sub.50 intensity HEK293-hCLDN18.1 Clone Number (μ g/mL) (Top MFI) Binding or not (+/-) ch-175D10 0.136 86407 – hu299B2-S32A-1 0.201 88800 – hu299B2-S32A-2 0.111 90420 – hu299B2-S32A-3 0.342 102785 – hu299B2-S32A-4 0.281 84871 – hu253C4-N31Q-1 0.190 64998 – hu253C4-N31Q-2 0.026 36841 – hu253C4-N31Q-3 0.030 42107 –

(131) Table 12 showed the affinity results for the humanized antibody of the present disclosure and the reference antibody ch-175D10 to HEK293-mCLDN18.2 cells and HEK293-hCLDN18.1 cells, respectively. Experimental results showed that: the humanized antibody of the present disclosure was the same as the reference antibody ch-175D10, both of which bound to the mCLDN18.2 antigen, wherein the binding of the humanized antibody to HEK293-mCLDN18.2 exhibited a mean maximum fluorescence intensity of 36841-102785 and a half binding concentration (BC.sub.50) of 0.026-0.342 μ g/mL, the binding of the reference antibody ch-175D10 to HEK293-mCLDN18.2 under the same reaction conditions exhibited a mean maximum fluorescence intensity of 86407 and a half binding concentration (EC.sub.50) of 0.136 μ g/mL, indicating that the binding of the humanized antibody to mCLDN18.2 comparable to that of the reference antibody. Moreover, the humanized antibody was the same as the reference antibody ch-175D10, neither binding to the hCLDN18.1 antigen.

Example 9 In Vitro Functional Assay of Humanized Antibodies

(132) A. Antibody-Dependent Cell-Mediated Cytotoxicity (ADCC)

(133) According to the method described in Example 5A, CHO-K1/hCLDN18.2 cells were used as target cells, NK cells transfected with 158V/V type Fc γ RIIIa gene (NK92/FcR γ 3a.158V/V) were used as effector cells, and the release of lactate dehydrogenase (LDH) in cell was detected by the cytotoxicity assay kit (Roche) and used as an indicator of cell killing effect.

(134) The percentage of cell lysis caused by the ADCC effect was calculated using the following formula: % cell lysis=100% $\times$ (sample release-target cell/effector cell mixed release)/(maximum release-target cell release), wherein the maximum release was the absorbance value produced in the wells of target cells treated with Triton X-100, the target cell/effector cell mixed release was the absorbance value produced in the wells of target cells and effector cell mixture, and the target cell release was the absorbance value produced in the wells containing only target cells, the sample release was the absorbance values produced in the wells of a humanized antibody, target cells, and effector cells mixture, and EC50 and maximal lysis were calculated by GraphPad software, and the results were shown in Table 13.

(135) TABLE-US-00013 TABLE 13 ADCC activity of humanized anti-CLDN18.2 antibodies Clone Number Maximum lysis (%) EC.sub.50 (μ g/ml) ch-175D10 51.42/49.15 0.047/0.039 hu299B2-S32A-1 57.89 0.022 hu299B2-S32A-2 68.08 0.028 hu299B2-S32A-3 69.09 0.022 hu299B2-S32A-4 63.23 0.024 hu253C4-N31Q-1 54.85 0.024 hu253C4-N31Q-2 58.80 0.037 hu253C4-N31Q-3 70.26 0.038

(136) Table 13 and FIGS. 34-35 showed ADCC results for the humanized antibody of the present disclosure and the reference antibody ch-175D10 on CHO-K1/hCLDN18.2 cells. Experimental results showed that the maximal ADCC effect of the humanized antibody of the present disclosure on CHO-K1/hCLDN18.2 cells was 54.85%-70.26%, and the ADCC effect of the reference antibody ch-175D10 was about 50% under the same reaction conditions. The concentration of the humanized antibody of the present disclosure that produces 50% ADCC effect (EC.sub.50) was

0.022-0.038 µg/mL, and the concentration of ch-175D10 that produces 50% ADCC effect (EC.sub.50) under the same reaction conditions was 0.040 µg/mL. The above results demonstrated that the humanized antibodies of the present disclosure were comparable to the reference antibody ch-175D10 in ADCC activity.

(137) B. Complement-Dependent Cytotoxicity (CDC)

(138) According to the method described in Example 5B, Cell viability was measured by CellTiter-Glo® chemiluminescent cell viability assay kit (Promega) using CHO-K1/hCLDN18.2 as target cells and pooled normal human serum (PNHS) as complement source, and the results were read using PHERAstar Plus software.

(139) The cell lysis rate caused by the humanized antibody in the CDC assay was calculated using the following formula:

$$\% \text{ cell lysis} = 100\% \times (1 - (\text{test wells} - \text{serum control wells}) / (\text{cell} + \text{serum wells} - \text{serum control wells}))$$

(140) The experimental controls are: the serum control wells: only serum (i.e., 30 µL buffer+10 µL diluted serum). Cells+serum wells: serum was added to wells of CHO-K1/hCLDN18.2 cell suspension (i.e., 20 µL cell suspension+10 µL buffer+10 µL diluted serum). Test wells: serum and chimeric antibody were added to wells of CHO-K1/hCLDN18.2 cell suspension (i.e., 20 µL cell suspension+10 µL antibody+10 µL diluted serum).

(141) The EC.sub.50 and maximum lysis were calculated using GraphPad software, and the results were shown in Table 14.

(142) TABLE-US-00014 TABLE 14 CDC activity of humanized anti-CLDN18.2 antibodies

Clone Number	Maximum lysis (%)	EC.sub.50 (µg/ml)	ch-175D10	87.76	1.549	hu299B2-S32A-1	99.15	0.125	hu299B2-S32A-2	99.74	0.129	hu299B2-S32A-3	99.83	0.092	hu299B2-S32A-4	99.87	0.123	hu253C4-N31Q-1	96.29	0.175	hu253C4-N31Q-2	97.51	0.149	hu253C4-N31Q-3	96.11	0.128
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(143) Table 14 and FIGS. 36-37 showed CDC results for the humanized antibody of the present disclosure and the reference antibody ch-175D10 on CHO-K1/hCLDN18.2 cells. Experimental results showed that: the maximal CDC effect of the humanized antibody of the present disclosure on CHO-K1/hCLDN18.2 cells was 96.11%-99.87%, and the maximal CDC effect of the reference antibody ch-175D10 was 87.76% under the same reaction conditions. The concentration of the chimeric antibody of the present disclosure that produces 50% CDC effect (EC.sub.50) was 0.092-0.175 µg/mL, and the concentration of ch-175D10 that produces 50% CDC effect (EC.sub.50) under the same reaction conditions was higher than 1 µg/mL. It can be seen that the CDC activity of all the humanized antibodies was stronger than that of the reference antibody.

(144) The embodiments of the present disclosure described above are intended to be merely exemplary, and equivalents of numerous specific compounds, materials, and operations may be recognized or determined by one skilled in the art without undue experimentation. All such equivalents are intended to be within the scope of the present disclosure and are encompassed by the claims.

Claims

1. An anti-CLDN18.2 antibody or antigen-binding fragment thereof, comprising a heavy chain variable region and a light chain variable region, the heavy chain variable region comprising an HCDR1, an HCDR2, and an HCDR3 and the light chain variable region comprising an LCDR1, an LCDR2, and an LCDR3 selected from the group consisting of: (1) SEQ ID NOs: 37, 38, 39, 86, 93, 97; (2) SEQ ID NOs: 40, 41, 42, 87, 94, 98; (3) SEQ ID NOs: 43, 41, 44, 88, 93, 99; (4) SEQ ID NOs: 45, 46, 47, 87, 95, 100; (5) SEQ ID NOs: 37, 48, 39, 88, 93, 97; (6) SEQ ID NOs: 49, 50, 51, 88, 93, 101; (7) SEQ ID NOs: 49, 52, 51, 89, 93, 102; (8) SEQ ID NOs: 53, 54, 55, 88, 93, 100; (9) SEQ ID NOs: 56, 57, 58, 90, 93, 103; (10) SEQ ID NOs: 59, 60, 61, 91, 96, 104; (11) SEQ ID NOs: 62, 63, 64, 88, 93, 98; (12) SEQ ID NOs: 65, 66, 67, 92, 93, 105; (13) SEQ ID NOs: 68, 69, 70, 88, 93, 106; (14) SEQ ID NOs: 71, 72, 73, 88, 93, 107; (15) SEQ ID NOs: 74, 75, 76, 88, 93,

106; (16) SEQ ID NOs: 77, 78, 79, 87, 93, 108; (17) SEQ ID NOs: 80, 81, 82, 88, 93, 109; (18) SEQ ID NOs: 83, 84, 85, 88, 93, 110; (19) SEQ ID NOs: 71, 72, 73, 112, 93, 107; (20) SEQ ID NOs: 71, 72, 73, 113, 93, 107; (21) SEQ ID NOs: 83, 84, 85, 111, 93, 110; (22) SEQ ID NOs: 83, 84, 85, 112, 93, 110; and (23) SEQ ID NOs: 83, 84, 85, 113, 93, 110; wherein the HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, and LCDR3 are arranged in the order of HCDR1, HCDR2, HCDR3, LCDR1, LCDR2, LCDR3 in each group.

2. The anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, comprising at least 80% to 100% sequence identity to a heavy chain variable region and a light chain variable region of any one of the groups consisting of: (1) SEQ ID NOs: 1 and 2; (2) SEQ ID NOs: 3 and 4; (3) SEQ ID NOs: 5 and 6; (4) SEQ ID NOs: 7 and 8; (5) SEQ ID NOs: 9 and 10; (6) SEQ ID NOs: 11 and 12; (7) SEQ ID NOs: 13 and 14; (8) SEQ ID NOs: 15 and 16; (9) SEQ ID NOs: 17 and 18; (10) SEQ ID NOs: 19 and 20; (11) SEQ ID NOs: 21 and 22; (12) SEQ ID NOs: 23 and 24; (13) SEQ ID NOs: 25 and 26; (14) SEQ ID NOs: 27 and 28; (15) SEQ ID NOs: 29 and 30; (16) SEQ ID NOs: 31 and 32; (17) SEQ ID NOs: 33 and 34; (18) SEQ ID NOs: 35 and 36; (19) SEQ ID NOs: 27 and 115; (20) SEQ ID NOs: 27 and 116; (21) SEQ ID NOs: 35 and 117; (22) SEQ ID NOs: 35 and 118; (23) SEQ ID NOs: 35 and 119; (24) SEQ ID NOs: 120 and 121; (25) SEQ ID NOs: 120 and 123; (26) SEQ ID NOs: 120 and 124; (27) SEQ ID NOs: 122 and 121; (28) SEQ ID NOs: 125 and 126; (29) SEQ ID NOs: 125 and 127; and (30) SEQ ID NOs: 128 and 126.

3. The anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, which is a murine antibody, a chimeric antibody, or a humanized antibody.

4. The anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, further comprising a heavy chain constant region and/or light chain constant region, optionally wherein the heavy chain constant region comprises an FC or a variant Fc.

5. A conjugate formed by coupling the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1 to a capture label or a detection label, the detection label comprising radionuclides, luminescent substances, colored substances, or enzymes.

6. A multispecific antibody, wherein one antigen-binding domain comprises the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1.

7. The multispecific antibody of claim 6, which is a bispecific antibody.

8. An antibody-drug conjugate, comprising the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, the antibody-drug conjugate being formed by antibody-linker-toxin interconnections.

9. A chimeric antigen receptor, comprising an extracellular recognition unit comprising the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1.

10. A nucleic acid encoding the anti-CLDN18.2 antibody or antigen binding fragment thereof claim 1.

11. A recombinant vector comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1.

12. A host cell comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1 being integrated into the genome or comprising a recombinant vector comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1.

13. A method of preparing the anti-CLDN18.2 antibody or antigen binding fragment thereof of claim 1, comprising: culturing one or more host cells comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof according to claim 1, or culturing one or more host cells comprising a recombinant vector comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof according to claim 1 under suitable conditions and purifying the expression products from the cells.

14. A pharmaceutical composition comprising an effective amount of the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, or comprising an effective amount of a

multispecific or bispecific antibody wherein one antigen-binding domain of the multispecific or bispecific antibody comprises the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, or comprising an effective amount of an antibody-drug conjugate comprising the anti-CLDN18.2 antibody or antigen binding fragment thereof of claim 1, or comprising an effective amount of a chimeric antigen receptor comprising an extracellular recognition unit comprising the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, or comprising an effective amount of a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof of claim 1, or comprising an effective amount of a recombinant vector comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof according to claim 1, or comprising an effective amount of a host cell comprising a recombinant vector comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof according to claim 1, or comprising an effective amount of a host cell comprising a nucleic acid encoding the anti-CLDN18.2 antibody or antigen-binding fragment thereof according to claim 1 being integrated into the genome the pharmaceutical composition further comprises a pharmaceutically acceptable carrier or one or more additional therapeutic agents.

15. A method of inducing cell death of CLDN18.2-expressing cells, comprising contacting the cells with the pharmaceutical composition of claim 14.

16. A method of treating a disease associated with expression of CLDN18.2 in a subject, comprising administering to a subject in need thereof the pharmaceutical composition of claim 14, optionally wherein the method further comprises administering to the subject in need an additional therapeutic agent.

17. The method of claim 16, wherein the disease is a tumor; and wherein the additional therapeutic agent comprises one or more selected from the group consisting of: chemotherapeutic agents, cytotoxic agents, radiotherapeutic agents, cancer vaccines, anti-neoplastic agents, targeted anti-cancer agents, anti-angiogenic agents, biological response modifiers, cytokines, hormones, anti-metastatic agents, immunotherapeutic agents, oncolytic viruses, and protease inhibitors.

18. The method of claim 16, wherein the disease is gastric cancer, esophageal cancer, intestinal cancer, pancreatic cancer, nephroblastoma, lung cancer, ovarian cancer, colon cancer, rectal cancer, liver cancer, head and neck cancer, chronic myelogenous leukemia, or gallbladder cancer.
